

**MINISTRY OF HIGHER AND SECONDARY SPECIAL
EDUCATION OF THE REPUBLIC OF UZBEKISTAN**

SAMARKAND STATE UNIVERSITY

**METHODOLOGICAL MANUAL
FOR PRACTICAL CLASSES ON
*BOTANY. SYSTEMATICS OF HIGHER PLANTS***

(for the second year students of Biological Faculty)

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“Methodological manual for practical classes on Botany. Systematics of Higher Plants” for conducting laboratory classes was developed according to the academic curriculum approved in 2020 for 5140100-Biology educational field. This methodological manual includes classification of plants, their morphological and anatomical characteristics, vegetative, sexual and asexual reproduction and distribution. Each class was illustrated with photographs and figures of life cycles of the target plants. The translation of the common botanical terms are given in Uzbek and Russian languages.

The current **“Methodological manual for practical classes on Botany. Systematics of Higher Plants”** can be used by students studying biology and ecology, in addition to the students who are taught Botany in higher educational institutions of the Republic of Uzbekistan.

Prepared

T.F.Rajabov

M.G.Nosirov

A.K.Akhmedov

Editor in Chief

Professor **Kh.K.Haydarov**

Reviewed

Assoc. Professor **Z.B.Nomozova**

Assoc. Professor **E.I.Hamdamova**

Professor **Ravinder Gargesh**

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INTRODUCTION

This methodological manual is prepared for teaching second-year students the course of Botany. Systematics of Higher Plant which is designed to provide hand-on practical classes. The manual is based on the National Program of the Republic of Uzbekistan on Training the Specialist and is prepared in accordance with the approved curriculum in the field of Biology 5140100 in the 2020 academic year. Many topics are based on local materials and reflect the phylogenetic relationship between the systematic taxa being studied. The sequence of study materials in the text is based on the latest developments in the systematics of higher plants and is intended to study plant kingdom of higher plants in 9 divisions. The manuals for the practical classes are available in Uzbek and Russian languages, reflecting the new system of higher plants. However, the textbooks in English language are not available for the students, and with that in mind, it was necessary to write this manual using the recent textbooks published in English language. The preparation of this manual is based on the classical literature published by authors such as L.I.Lotova (2018), V.M.Basov, T.V.Efremov (2018), and O.V.Korovkin (2018). Along with these works, the manual was methodologically enriched by textbooks in English such as “A Photographic Atlas for the Botany Laboratory” (Kent M. Van De Graaff, Samuel R. Rusgforth, John L. Crawley, 2004), “An introduction to plant structure and development” (Beck, 2005), “Plant biology” (Lack, A.J. and D.E. Evans, 2001). The classification of plants is based on the work of U.P. Pratov, T. Odilov - "Modern system of higher plant families of Uzbekistan".

The purpose, tasks, necessary equipment, recommendations on the steps in work are given in each practical class, which is aimed at consolidating the knowledge gained on a particular topic and at developing the students' ability to conduct practical experiments on the topics independently.

The methodological manual for practical classes on higher plant taxonomy is intended for students and teachers of biology, ecology of

Samarkand State University, but it can also be used by students and teachers of other higher educational institutions of the Republic to teach classes on Botany.

This handbook provides translations of about 350 botanical terms commonly used in classrooms through the Uzbek, English, and Russian languages. The botanical terms are given in alphabetical order at the end of the manual. Species names of the plants in the classes are given in common English and Latin, but throughout the text scientific Latin names are used frequently.

We acknowledge the valuable methodological comments provided by Dr. Nomozova Z.B., Dr. Hasanov M.A. and Dr. Umurzakova Z.I. for the improvement of this manual.

DIVISION *BRYOPHYTA* – *Mosses*

Bryophyte is the traditional name for any nonvascular seedless plant—namely, any of the mosses, hornworts, and liverworts. The scientific study of bryophytes is known as bryology. It is believed that the division Bryophyte evolved from green algae. Fossil evidence indicates that bryophytes are the oldest of the four major plant groups (Bryophytes, Pteridophytes, Gymnosperms, and Angiosperms). The first bryophytes probably appeared in the Ordovician period, about 450 million years ago. However, because of the lack of lignin and other resistant structures, bryophyte fossil formation is improbable and the fossil record is also poor, thus restoring of their full phylogenesis is difficult.

The Bryophyta division includes almost 50 000 species. Among the higher plants they are the second most diverse division in species after angiosperms. Bryophytes are the simplest and the most primitive plants. This is the first group of plants to invade the land, though they require water for their fertilisation. Prominent bryophytes characteristics are the absence of true root stems and leaves. In a bryophyte, all the conspicuous vegetative organs, including the photosynthetic leaf-like structures, the thallus, stem, and the rhizoid anchors the plant to its substrate. Furthermore, rhizoids perform the function of roots, essentially anchoring the plants into the surface, though, they do not absorb nutrients like traditional plant roots.

They generally lack lignin and do not have actual tracheids (xylem cells specialized for water conduction). Rather, water and nutrients circulate inside specialized conducting cells. Although the term non-tracheophyte is more accurate, bryophytes are commonly called non-vascular plants.

The bryophytes show an **alternation of generations** between the independent gametophyte generation (that produces the sex organs and sperm and eggs) and the dependent sporophyte generation (which produces the spores). The sporophyte is barely noticeable. Thus, the gametophyte is the dominant and most familiar form; the sporophyte

appears for only a short period. The gametes formed by bryophytes swim with a flagellum. The sporangium, the multicellular sexual reproductive structure, is present in bryophytes but is absent in the majority of algae. The sporophyte embryo also remains attached to the parent plant, which protects and nourishes it. This is a characteristic of land plants. They do not have leaves rather they have leaf like scales that contain chloroplasts for photosynthesis. Bryophytes lack the true stem. Rhizoids are unicellular. These rhizoids perform the function of fixation and absorption as roots but they are not regenerated as true root.

Bryophytes are distributed in a variety of ecological environments. They are found in areas ranging from the tropical and subtropical to the cold tundra. Another characteristic feature of bryophytes in the developmental cycle is the growth of a protonema after the maturation of mature spores.

The protonema has a fibrous structure in most bryophyte species. Only the sphagnum and andrea bryophytes are platelets in the early phase of the gametophyte. In liverworts, the protonema consists of a short filament. It consists of one or more cells. Its apical cell divides longitudinally into two from which a platelet or stem gametophyte develops.

Among the bryophytes, the protonema of the funaria (*Funaria hydrometrica*) is well studied. Its spores grow rapidly under favorable conditions and form a filament. After that, the rhizoid develops. The green fibrous protonema is called chloronema. After branching of the filament, it passes into a stage called caulonema. This is the final stage of the protonema, which contains gametophore buds and rhizoids. This filamentous protonema of bryophyte is very similar in appearance to green algae. Therefore, it is believed that bryophytes originated from green algae. However, there is no solid evidence to support this view. According to the current classifications, the division of bryophyte plants is divided into three classes: the liverworts (*Hepaticopsida*), the hornworts (*Anthocerotopsida*), and the mosses (*Bryopsida*).

1-Class: **Liverworts** - *Marchantiopsida*, *Hepaticopsida* (s)

Orders: a) *Marchantiales*

b) *Jungermanniales*

v) *Sphaerocarpaceae*

2-Class: **Hornworts** - *Anthocerotopsida*

Order: *Anthocerotales*

3-Class: **Mosses** - *Bryopsida*, *Musci*

Orders: a) *Sphagnales*

b) *Andreales*

v) *Bryales*

In conclusion, the following common features of the bryophyte division can be given: bryophyte as known as an amphibious zone; bryophyta grows on tree trunks, rocks and in moist area of the soil; bryophytes are the only embryophytes whose life history includes a dominant gametophyte (haploid) stage; they are an ancient and diverse group of non vascular plants; they comprise three main taxonomic groups - mosses, liverworts and hornworts which have evolved quite separately.

1-CLASS

TOPIC: STRUCTURE AND DEVELOPMENT OF *MARCHANTIA POLYMORPHA*

Division: *Bryophyta*

Class: *Hepaticopsida*, *Marchantiopsida*

Order: *Marchantiales*

Family: *Marchantiaceae*

Species: *Marchantia polymorpha*

Purpose of the class: to study the structure, development stages and classification of *Marchantia polymorpha*.

Objectives of the class: to teach students herbarium specimens, and to inform them about live materials and classification of the target plant; to strengthen the student's knowledge of the subject; to be able to answer the teacher's questions.

Demonstration materials and equipments: powerpoint slides; animations, online materials. Sketch of reproduction cycle of *M. polymorpha*, herbarium specimens, plant exhibits, prepared microscope slides (antheridium, archegonium, sporogonium), microscopes, magnifying glass, microscope slides, cover slips, preparation needles, water dropper (pipette).

LEARNING PROCEDURE

Students will get to know the herbarium specimens and exhibits of *Marchantia polymorpha* and learn about their biological characteristics using the teaching materials and books.

GAMETOPHYTE: A gametophyte, or sexual generation, is a plant that grows from spores.

Take a **capsule** of *Marchantia* and use a needle to remove some of the spores. Microscope slide of spores is prepared by placing the spores on a microscope slide and it will be viewed first through a small microscope objective lens and then through a large one. The microscope shows round spores and **spring-shaped elaters**. Skin and a nucleus of spores, and the changing of elaters by elongation can be observed. Such changes in elaters confirms disperse of spores from the capsule of *Marchantia*. A sketch of spores and spring-shaped elaters is drawn in the album.

The protonema is a plate-like, algae-like initial plant and it develops from the spores when moist soil is available and the temperature is high enough. When these **growths (prothallus)** appear on the soil surface, they turn green, grow as a result of cell division, and form a dichotomically branched young plant of *Marchantia* (Fig. 1).



Figure 1. *Marchantia polymorpha* and its gametophyte thallus

Using the herbarium specimen, the external-morphological structure of the *Marchantia* thallus will be familiarized with a **magnifying glass**. The students should pay attention to the branching type of thallus, the thallus with wider discs (**antheridial thallus or antheridiophore**) on the stalk, and stellar longer light structures (**archegonic thallus - archegoniophores**) on other thalli. The upper and lower sides of the thallus are carefully examined, and it is observed that they differ in color and outgrowth, especially in the presence of **gemma cups** (on top of the thallus), rhizoids, and amphigastria (underneath the thallus).

The microscope slide is prepared from the cross section of the thallus, the lower and upper epidermis, air chambers, assimilation cells, stomata are found through a small microscope lens and are drawn on an album sheet.

The growing point of *Marchantia* thallus and its location will be found and observed, and it will also be drawn down in the album.

From the herbarium specimen, the microscope slide is prepared from a longitudinal section of the apical part of **antheridial and archegonic stalk** and viewed through a small microscope lens. The tissue in the thallus is compared to the vascular tissue of the stalk. The antheridial and archegonic stalk are compared and studied. Antheridium

and archegonia are found, and attention is paid to their shape. The number of layers of their wall cells can be observed through a small and a large microscope lens. Antheridial and archegonic stalk, sexual organs are drawn on the album page. **Biflagellate spermatozoon** in the antheridium will be moved by water and will find **egg cells (ovum)** of the archegonium in order to fertilize one, and then a zygote is formed.

Below in the Fig.2, students will get familiarized with the development cycle of *M. polymorpha*.

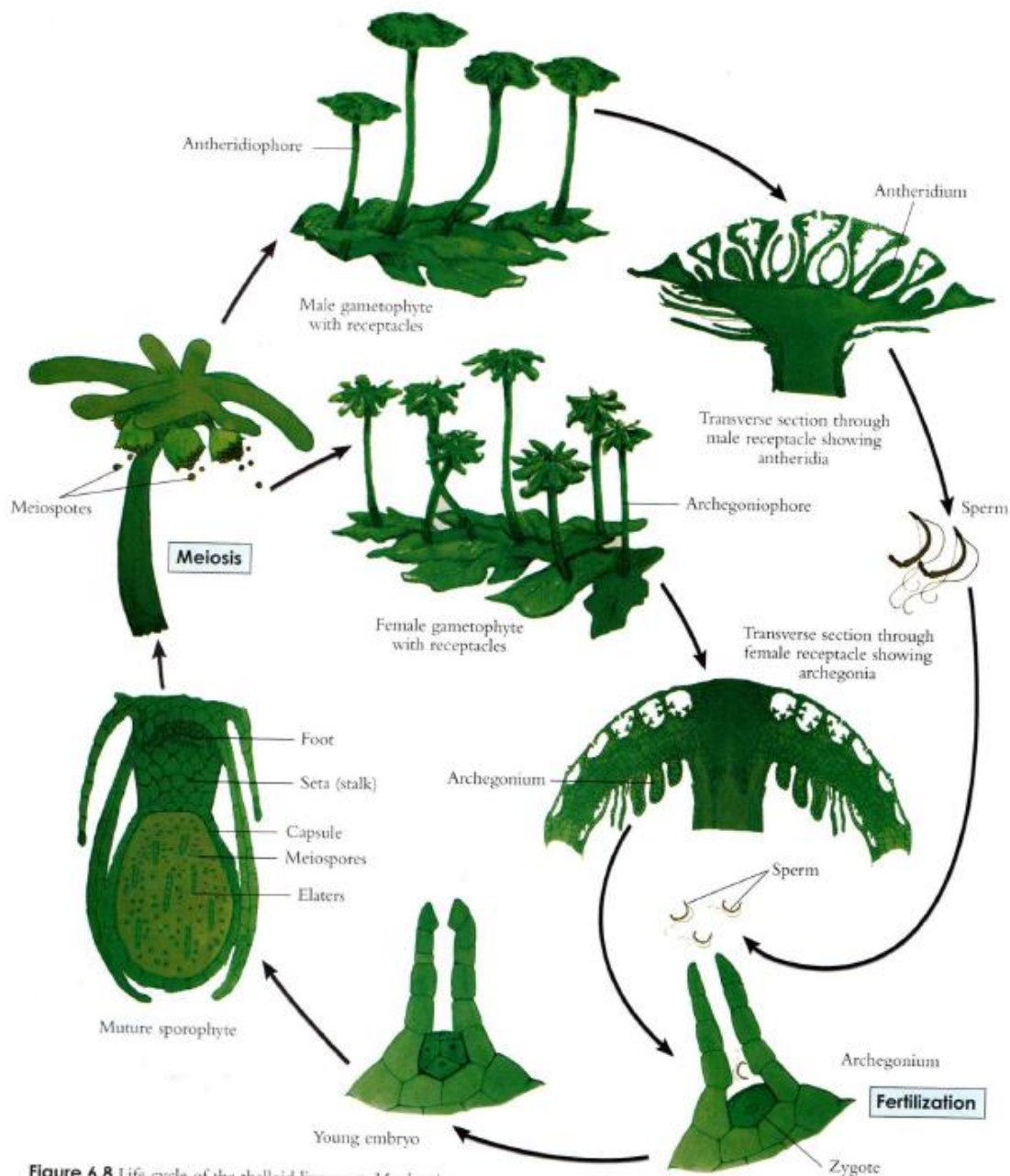


Figure 6.8 Life cycle of the thalloid liverwort, *Marchantia*.

Figure 2. Development cycle of *Marchantia polymorpha*

SPOROPHYTE: A sporophyte is formed by the division of a zygote cell. In *Marchantia*, the sporogonium, namely **capsule** and its **seta** forms the sporophyte. Initially, the shape, size, and parts of the sporogonium are examined, and rapid removal of **calyptra** is observed. The capsule is examined under a microscope to determine how many layers of cells are in the wall. The spores inside the capsule are formed by archeospore cells. Some cells will not get divided and become elaters. Everything that can be seen under a microscope will be recorded in an album.

Self Testing Questions:

1. What plant structure and reproduction characteristics of Bryophytes make them different from higher plants?
2. What is the structure of gametophyte of *Marchantia polymorpha*?
3. What is the protonema of Bryophytes?
4. How does protonema germinate and what specific condition is necessary for germination?
5. What are the functions of rhizoids?
6. What does *M. polymorpha* sporophyte consist of?
7. What generation of *M. polymorpha* holds stalks with disc?
8. What is the structure of antheridiophore and archegoniophore in *Marchantia*?

2-CLASS

TOPIC: THE STRUCTURE AND DEVELOPMENT OF *FUNARIA HYGROMETRICA* AND *POLYTRICHUM COMMUNE*

Division: *Bryophyta*

Class: *Bryopsida* or *Musci*

Subclass: *Bryidae*

Order: *Bryales*

Species: *Funaria hygrometrica*

Species: *Polytrichum commune*

Purpose of the class: to study the structure, development stages and classification of *Funaria hygrometrica* and *Polytrichum commune*.

Objectives of the class: to teach herbarium specimens to students, and to inform them about live materials and classification of the target plant; to strengthen the student's knowledge of the subject; to be able to answer the teacher's questions.

Demonstration materials and equipments: powerpoint slides; animations, online materials, electronic educational instructions and manual. Herbarium specimens of *F. hygrometrica* and *P. commune*, plant exhibits, prepared microscope slides (antheridium, archegonium, sporogonium), microscopes, magnifying glass, microscope slides, cover slips, preparation needles, water dropper (pipette), sketch of reproduction cycle.

LEARNING PROCEDURE

Students will get to know the herbarium specimens and exhibits of *Funaria hygrometrica* and *Polytrichum commune* and learn about their biological characteristics using the teaching materials and books.

GAMETOPHYTE: The gametophyte is a sexual generation formed by the growth and development of spores. Spores, on the other hand, grow in sporogonium. Spores are separated from the capsule of *F. hygrometrica* and *P. commune*, using this the microscopic slide is prepared and viewed through a small and large microscope lens, attention will be paid that there are no elaters. The microscopic view of the spores structure will be drawn. When the spores fall under favorable conditions, they form a fibrous, branched protonema, which is the beginning of the gametophyte. The difference of protonema from fibrous algae will be established. Buds are found in the protonema. It is believed that they form the thallus of **true mosses** with leaves and stems. A picture of the protonema and its buds will be drawn.

By looking at the growing *Funaria*, the color of the lower and upper parts of the stem is determined and compared. Rhizoids between soils and their structure are studied using magnifiers.

To study the internal structure of the stems and leaves of the true mosses, the microscopic slide is prepared by cutting off the leaf-covered part of the stem and examining it under a small lens of a microscope, the dark colored cells at the edge of the stems forms the bark and the central cells form **vascular bundles**. The color and thickness of the bark layers are compared by comparing two species. Attention will be drawn to the fact that the outer layer of cells forms the mechanical tissue, while the inner layer forms the main tissue. Examination of the vascular bundles of both species reveals that there are more dead cells in the central part than in the periphery. This is especially true in *P. commune*. A diagram of the cross section of the stem need to be drawn on an album sheet.

Under the microscope, take a look at the anatomical structure of the *F. hygrometrica* and *P. commune* leaves, and observe the location of the assimilation cells of the leaf. Attention should be paid to the **midrib** of *Funaria*, that its leaf consists of several layers of cells but the lateral “wing” is made up of a single layer of thin walled cells which are rich in chloroplasts. Its sketch should be drawn in an album.

F. hygrometrica and *P. commune* are **dioecious** and **unisexual** plants.

The sexual organs – the clusters of archegonia and antheridia, are located between the leaves at the apex (upper part) of the stem. In the alive or growing *Funaria*, archegonium and antheridium holding stems are compared. The growing *Funaria* can be obtained from the field (outside) in spring, in autumn and winter seasons from the greenhouses. Attention is paid to how they differ from each other. Attention is also paid to the **paraphyses** (multicellular hairs) which are located between the archegonia in the apex of the stem. The thickness of wall cell of archegonia is determined. A longitudinal section of the stem apex of an archegonic plant should be drawn.

The longitudinal section of the stem apex of the antheridial plant is also examined under a microscope, and the differences are identified (especially compared with the antheridia of *Marchantia*). The longitudinal section of the stem apex of the antheridial plant is also drawn. In true mosses the sperm cells or spermatozoon are mobile, and

an aqueous environment or water is essential for taking place of fertilization. Fertilization results in the formation of a zygote, and the division of the zygote indicates the start of sporophyte generation.

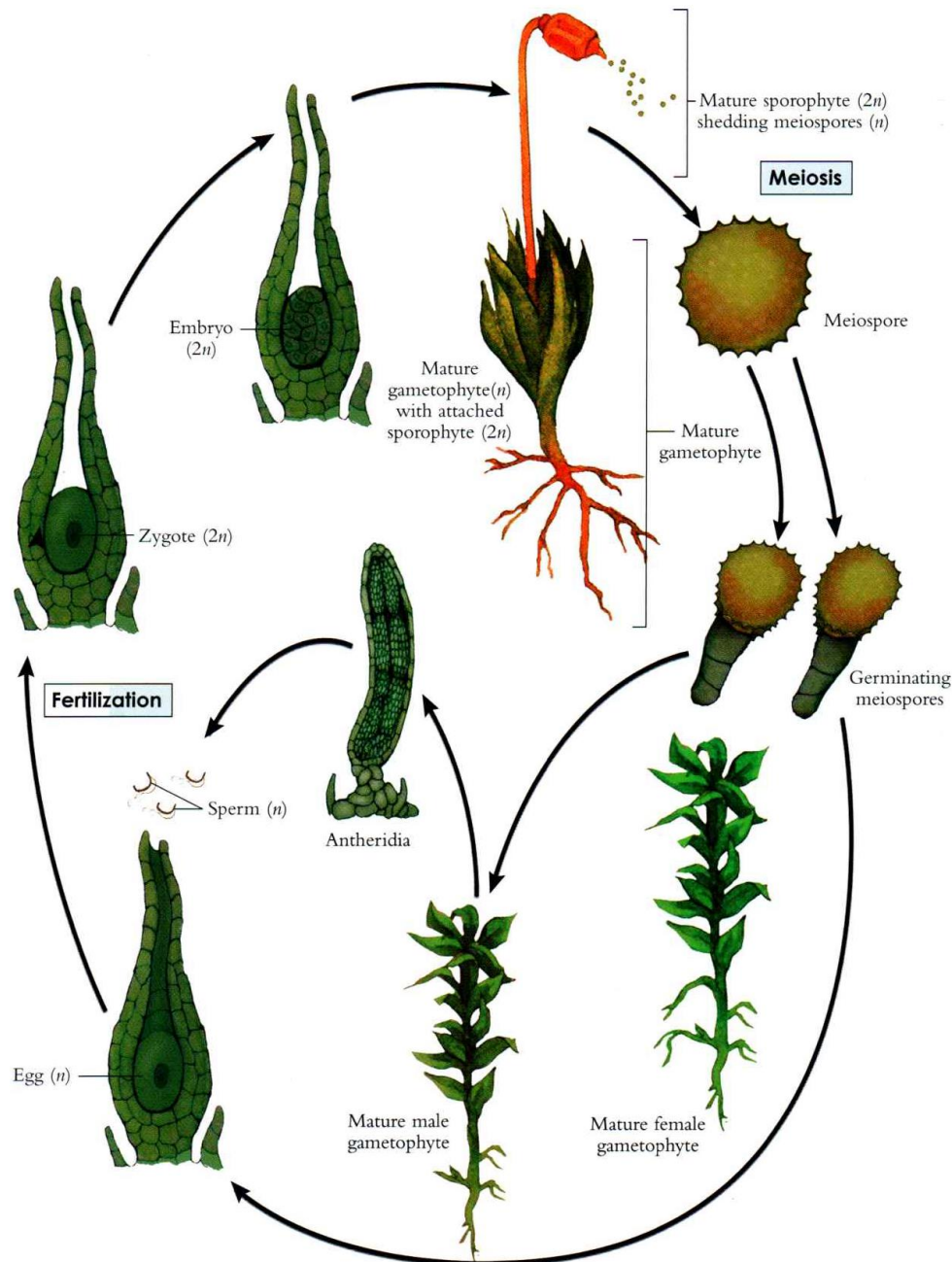


Figure 3. Alternation of generation of *Funaria hygrometrica*

SPOROPHYTE: The fertilization and development of an egg cell in archegonia produces a sporophyte. The sporophyte of the *F. hygrometrica* and *P. commune* consists of a sporogonium, namely capsule and seta. The microscope slide is prepared from the longitudinal

section of the capsule and observed under a microscope. In the microscope slide, the following parts can be noticed: apophysis, trabeculae, columella, sportes sac, annulus and operculum, peristome teeth and others. The capsules of the two species are compared. A picture of the structure of the longitudinal section of capsule is drawn on the album page. Using Figures 2 and 3, the development cycles are introduced, the sketch is drawn and the parts are recorded.

Self Testing Questions:

1. What does the gametophyte of *Polytrichum commune* consist of?
2. What does the sporophyte of *P. commune* consist of?
3. What does antheridium and archegonium consist of?
4. How does the reproduction cycle of *P. commune* take place?

DIVISION LYCOPODIOPHYTA – *Club mosses, quillworts and spike mosses*

This division is a first sporophyte dominant division in the evolution of higher plants. The representatives of this division are the oldest spore-bearing higher plants on Earth. Their fossils have been found in the layers of the Silurian period. At that time, there were many species of trees up to 30 meters tall among the club moss (*Lycopodium*). Remains of lepidodendrons and sigillaria orders were involved in the formation of coal. Its current inhabitants are club mosses (*Lycopodium*), spike mosses (*Selaginella*) and quillworts (*Isoetes*). Plants within the division *Lycopodiophyta* are small, green, leafy and have spores but no flowers. They are a little like mosses, and many contain the word ‘moss’ in their common names. However, *Lycopodiophyta* evolved later than plants in the *Bryophyta* division and have a fully formed **vascular system**, with phloem and xylem, they are the earliest vascular plants to have evolved.

They are true plants with stem and leaves and will always have roots. The stems are dichotomously branched.

They have simple leaf, and a bundle of water tubes which run through the leaves. Sporophylls are with single-sporangia and within them spores get matured. Spores can be of the same size (as the club mosses) or of a different size (like the spike mosses). Club mosses, like mosses, are highly spore-bearing plants that reproduce by alternation of sporophyte and gametophyte generation, as well as by vegetative organs. Unlike mosses, this division is dominated by asexual sporophyte generation. When the sporangia gets cracked or broken, the spores spread around. Under favorable conditions, they develop into a prothallus of **sexual phase** or gametophyte. The egg and sperm cells in the antheridia and the archegonia of the gametophyte phase ~~will~~ combine to form the zygote, and as a result of the zygote's continuous division, a new sporophyte, or asexual generation develops. The sporophyte grows independently of the gametophyte.

Almost 1200 species are known in the division *Lycopodiophyta*.

Classification of *Lycopodiophyta* division is as below:

1-class: *Lycopodiopsida*

Orders: 1. *Asteroxylinales*

2. *Protolpidodendrales*

3. *Lycopodiales*

2-class: *Jsoetopsida*

Orders: 1. *Lepidodendrales*

2. *Selaginellales*,

3. *Jsoetales*.

3-CLASS

TOPIC: STRUCTURE AND DEVELOPMENT OF *LYCOPODIUM CLAVATUM*

Division: *Lycopodiophyta*

Class: *Lycopodiopsida*

Order: *Lycopodiales*

Family: *Lycopodiaceae*

Species: *Lycopodium clavatum*

Purpose of the class: to study the structure, development stages and classification of *Lycopodium clavatum*.

Objectives of the class: to teach herbarium specimens to students, and to inform them about live materials and classification of the target plant; to strengthen the student's knowledge of the subject; to be able to answer the teacher's questions.

Demonstration materials and equipments: powerpoint slides; animations, online materials, electronic educational instructions and a manual. Herbarium specimens of *L. clavatum*, plant exhibits, prepared microscope slides (antheridium, archegonium, **sporoous capsule**), microscopes, magnifying glass, microscope slides, cover slips, preparation needles, water dropper (pipette), sketch of reproduction cycle.

LEARNING PROCEDURE

Students will get to know the herbarium specimens and exhibits of *L. clavatum* and learn about their biological characteristics using the teaching materials and books.

SPOROPHYTE. Lycopodium's asexual generation develops from an **embryo** which is formed by a fertilized egg cell, namely **zygote**. In this case, the set of chromosomes is always a diploid (2n) number. A sporophyte generation is an adult plant. From the available herbarium materials, the vertical and horizontal stems of *Lycopodium* are observed with their branching and additional roots. Leaf shapes and leaf arrangements on the stem, the number of spores in each seta on the top of the stem, and other morphological features of the plant are studied. Sporophylls in the sporous capsule and spore-forming sporangia in the sporophyll are studied using a magnifying glass (Fig. 4).

From the herbarium, the microscope slide of the cross-section of the *Lycopodium* stem is prepared or the prepared microscope slide is examined under a microscope. The type of **vascular bundle** in the cross-section, the location of the xylem and phloem are determined. The set of brown cells in the cortex is recognized as leaf traces. This illustrations, viewed under a microscope and a magnifying glass, is drawn on an album sheet.

Using a needle, a single sporophyll is separated from the sporous capsule and by a magnifying glass its shape and the location of the sporangium is determined. The prepared microscope slide of the longitudinal section of sporous capsule is viewed under a microscope and its structure is drawn on an album. Attention is paid to the shape of the sporangium and how it is attached to the sporophyll. The sporophyll and sporangium is drawn into album.

The sporangia will be broken with a needle on microscope slide, and the spores are seen first under a small lens and then under the large lens of microscope. The shape and size of the spores are determined to be the same or different. Draw the structures seen under the microscope in the album.

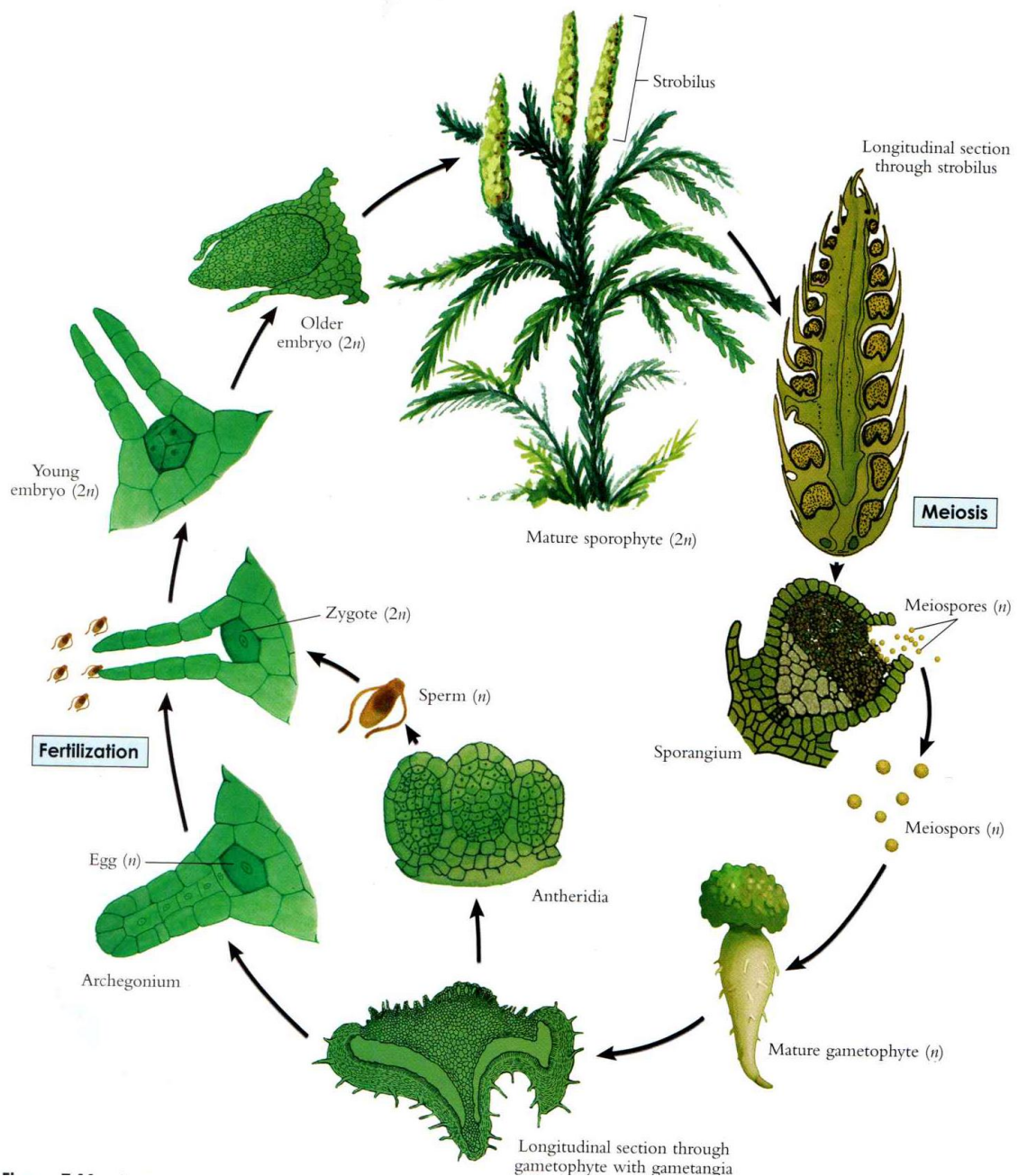


Figure 4. Development cycle of *Lycopodium clavatum*

GAMETOPHYTE: The creation formed by the germination of spores is gametophyte. The structure of the gametophyte is studied using the prepared microscope slide and compared with the picture of the prothallus in the table.

In gametophyte the emphasis is given on the presence of rhizoids in prothallus, symbiosis living with **fungal hyphae**, chlorophyll-free, monoecious bisexual, covered with epidermis on the outside, and the

central parts of the prothallus are composed of normal parenchymal tissues, and their diagram is to be drawn on the album sheet.

Fertilization takes place in an aquatic environment. It takes 15 to 18 years for a young sporophyte to emerge from the formed zygote. The structure and development cycle of *L. clavatum* in Fig. 4 will be studied and the picture needs to be drawn on the album.

Self Testing Questions:

1. What is the structure of gametophyte and sporophyte of *Lycopodium clavatum*?
2. Where and when the spores are developed?
3. What does the sporous capsule of *L. clavatum* consist of?
4. Where are the antheridium and archegonium located in the gametophyte?
5. What is the structure of the gametophyte of *L. clavatum*?
6. How many years are necessary for its sporophyte to mature?

DIVISION *PSILOTOPHYTA* – Whisk ferns

This division includes about 12 species. They consist of single class, order, family, and 2 genera: *Psilotum* and *Tmesipteris*.

Genus *Psilotum* includes 2 species. They are distributed in areas with tropical and subtropical climates - South Korea, South Japan, Hawaii and New Zealand. Representatives of the genus *Tmesipteris* (10 species) are distributed from Australia, Tasmania, New Zealand to the Philippine Islands, and partly in India. They grow as epiphytes, sometimes in rock crevices and in soils rich in soil organic matter.

Representatives of the division are one of the most ancient species of higher plants with a very simple structure. They have almost no economical importance. Only the common *Psilotum* is grown in Japan as an ornamental plant. Their underground parts are composed of dichotomically branched rhizomoids resembling rhizomes. They are covered with many rhizomes. It is similar to rhinophytes in this respect. The outer skin cells of the rhizomoids contain fungal hyphae that form endotrophic mycorrhiza. At the ends of some rhizomoids of the common species, detached buds are formed, from which new rhizomoids grow and reproduce vegetatively. From the rhizomoids of *Psilotum* species grow dichotomously branched shoot system or **aboveground parts**. They have no true roots.

The stems of psilots are 10-100 cm long, and in the *tmesipteris* 5-25 cm (sometimes up to 40 cm) long. In some species, the stem grows vertically or has an epiphytic stem hanging down. Representatives of the division are often found as epiphytes at the base of woody ferns. The stem of *Psilotum triquetrum* is triquetrous (or trihedral) and flattened in *P. flaccidum*. The stems of both species are dichotomously branched, but differ slightly in terms of the anatomical structure of the stem. The stem of the *Psilotum* is covered with epidermis on the outside. It has cuticles and stomata, with a three-layered cortical parenchyma inside, and its cells are rich in chlorophyll. In the center of the stem are ladder and spiral tracheids with an actinostel-type **bast layer**. The outer skin of the stem of the *tmesipteris* has no stomata. In them, the bast layer is well

developed, and the endoderm is less developed. The leaves of the psilotum are in the form of small coins. They have no leaf veins. The stem performs the function of assimilation. The leaves of the tmesipteris are wider and the **lanceolate tissue** is better developed.

At the ends of dichotomously branched sporophytes are formed synangium (a cluster of sporangia that have become fused is called a synangium). When the sporangia mature, they open in the form of a radial crack. Spores are the same size. From the spores grow a worm-shaped gametophyte. It is in a dichotomous form 1.8–2 cm long. The gametophyte contains rhizoids, which are underground, chlorophyll-free, and feed on saprophytes. Young gametophytes are similar to sporophytes in appearance, bisexual. The antheridia are spherical in shape, in which a number of spirally twisted spermatozoon are formed. They form an ovary after fertilizing the egg cell of the archegonia in the gametophyte. However, the embryo does not have a primary root, primary leaves. Later, a young sporophyte with dichotomous branches grows from the stalk. Leaves are formed in it. From the underground part, rhizomoids develop. Fossil representatives of psilotum have not been found. Psilotum is believed to have originated from rhinophytes.

4-CLASS

TOPIC: STRUCTURE AND DEVELOPMENT OF *PSILOTUM TRIQUETRUM*

Division: *Psilotophyta*

Class: *Psilotopsida*

Order: *Psilotales*

Family: *Psilotaceae*

Species: *Psilotum triquetrum*

Purpose of the class: to study the structure, development stages and classification of *Psilotum triquetrum*.

Objectives of the class: to teach students herbarium specimens, and to inform them about live materials and classification of the target plant; to strengthen the student's knowledge of ~~on~~ the subject; to be able to answer the teacher's questions.

Demonstration materials and equipments: powerpoint slides; animations, online materials. Sketch of reproduction cycle of *P. triquetrum*, herbarium specimens, plant exhibits, prepared microscope slides (antheridium, archegonium), microscopes, magnifying glass, microscope slides, cover slips, preparation needles, water dropper (pipette).

LEARNING PROCEDURE

Students will get to know the herbarium specimens and exhibits of *P. triquetrum* and learn about their biological characteristics using the teaching materials and books.

It is a small ancestor of a family of two genera called *Tmesipteris* (*Tmesipteris*) and *Psilotum* (*Psilotum*) (single species: *T. tannensis*). Occurs in Australia, New Zealand and Polynesia islands; *Psilotum* (two species called *P. triquetrum* and *P. flaccidum*) is distributed in all tropical and subtropical regions of both hemispheres. *Tmesipteris* is an epiphyte that grows on the bodies of woody **ferns**. The highly branched **rhizomes** of *tmesipteris* grows between the roots of the ferns, the leaf bases and in between ~~of~~ the decayed plant materials. From these rootstock protrude above-ground branches and hang on the body of ~~the~~ ferns. The above-ground branches are sometimes simple, sometimes dichotomous, and the stems are angular and covered with lanceolate hard leaves. Between the vegetative leaves at the top of the stem are sporophyll, which are divided into two and form large double sporangia. All spores are of the same size.

The gametophyte of *tmesipteris* is very similar in appearance to the gametophytes of *Lycopodium*. This gametophyte also grows underground and consists of a short, colorless **worm-like body** 2-3 cm long. Antheridia and archegonia appear in large numbers on the entire surface of the gametophyte. The shape of the antheridia is round; they

produce many spermatozoon. The archegonia are embedded in the gametophyte tissue with the abdomen and protrude with a short neck.

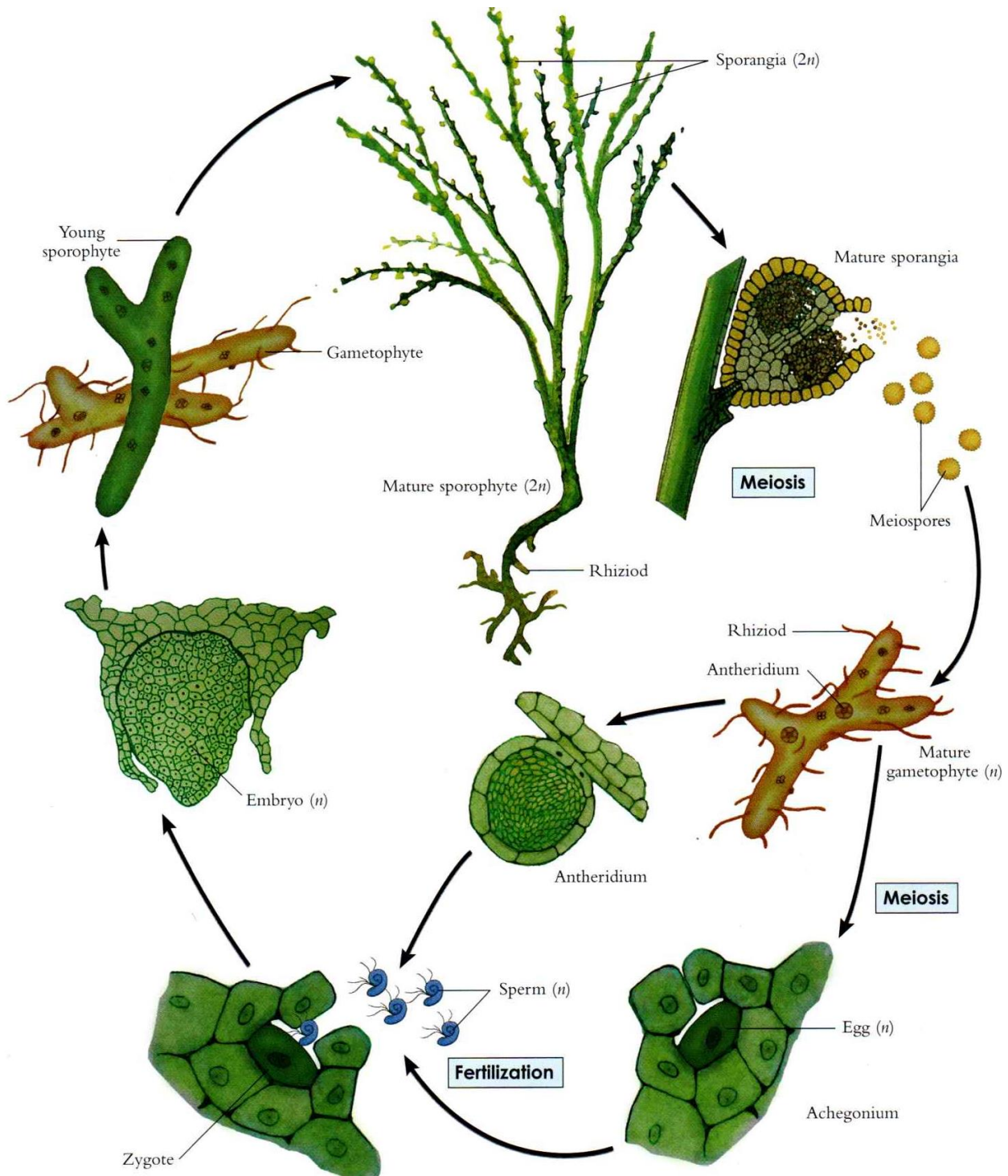


Figure 5. Development cycle of *Psilotum triquetrum*

Psilotum, which grows on the ground or in various trees as an epiphyte, has no roots and is a very branched rhizome. From the rhizome

emerges dichotomously branched, lanceolate-shaped, superficial branches covered with tiny sparse leaves resembling small excesses. At the top of the stems are large sporangia with two leaves at the base. Each sporangium is divided into three nests filled with spores. *Psilotum* spores are of the same size as *Tmesipteris* spores. The gametophyte of *Tmesipteris* is the same as the gametophyte of *Psilotum*.

Self Testing questions:

1. How is the sporophyte of *Psilotum* formed?
2. Where is the sporangium of *Psilotum* located?
3. What is the structure of *Tmesipteris*'s sporangia?
4. What is the structure of *Tmesipteris*'s gametophyte?

DIVISION *EQUISETIOPHYTA* - *Horsetail*

Equisetophyta, small division of the plant kingdom consists of the plants commonly called horsetails and scouring rushes. *Equisetum*, the only living genus in this division, is descended evolutionarily from tree-sized fossil plants. The plants, which generally grow in moist places, have roots and ribbed green stems, the surface of which is impregnated with silica crystals. The leaves do not have the ability to photosynthesize.

According to paleobotanical data, horsetail originated in the Devonian period of the Paleozoic era, with the peak of the Carboniferous period. The first tree species later became extinct, as did other highly spore-bearing plants. At present, only herbaceous horsetails with rhizomes are preserved.

There are 32 species of horsetails that can be found all over the world except Australia and New Zealand. There are 12 species in the CIS (Commonwealth of Independent States). *Equisetum giganteum* grows up to 10-12 meters tall in South America. There is a space in the center of the stem of the horsetails, which is called the central air space. The representatives of the division have an underground crawling rhizome, which store starch reserves. From the rhizome joints grow thin, dark extra roots. The rhizome is a organ of vegetative reproduction.

There are two species of *Equisetum* in the flora of Uzbekistan: *Equisetum arvense* and *Equisetum ramosissimum*.

In horsetails, as in the Lycopodiophyta, the sporophyte generation is predominant during the reproductive period. There are two types of stems in horsetail. The stems differ greatly in morphological structure and function. Spring stems grow in early spring, yellowish, dark brown, in which sporophyll spikes are formed. Sporophylls are thyroid-shaped, deformed leaves. They contain 5-13 sporangia. Sporophylls are located perpendicular to the axis of the spike using a band. Spores mature in sporangia. When the spore is exposed to favorable conditions, a gametophyte develops from it. Gametophytes are unisexual.

The ovum of an archegonia fuses with one of the sperm to form a zygote. From the zygote's tissue, an adult sporophyte is formed.

Classification of Equisetiophyta:

1-Class - *Sphenopsida*

Orders: a) *Hyeniales*,

b) *Sphenophyllales*

2-Class: *Equisetopsida*

Orders: a) *Calamitales*,

b) *Equisetales*.

5-CLASS

TOPIC: STRUCTURE AND DEVELOPMENT OF *EQUISETUM ARVENSE*

Division: *Equisetiophyta*

Class: *Equisetiopsida*

Order: *Equisetales*

Family: *Equisetaceae*

Species: *Equisetum arvense*

Purpose of the class: to study the structure, development stages and classification of *Equisetum arvense*.

Objectives of the class: to teach students herbarium specimens, and to inform them about live materials and classification of the target plant; to strengthen the student's knowledge of the subject; to be able to answer the teacher's questions.

Demonstration materials and equipments: powerpoint slides; animations, online materials. Sketch of reproduction cycle of *E. arvense*, herbarium specimens, plant exhibits, prepared microscope slides (antheridium, archegonium), microscopes, magnifying glass, microscope slides, cover slips, preparation needles, water dropper (pipette).

LEARNING PROCEDURE

Students will get to know the herbarium specimens and exhibits of *Equisetum arvense* and learn about their biological characteristics using the teaching materials and books.

SPOROPHYTE: *E. arvense* has separate sterile non-reproductive (with assimilative stems) and fertile spore-bearing (generative) stems. Generative stem is called spring stems, and assimilative non-reproductive stem is called summer stems. Asexual sporophyte generation is predominant in *E. arvense* (Fig. 7).

From the herbarium materials of the *E. arvense*, the appearance of the stem, the location of the stem and lateral branches, the shape of the leaves, the order of arrangement, color, where the roots and nodules are formed are studied with the help of a magnifying glass. The spring stem is observed, and attention is paid to what the top part of it ends. When the spring stem forms spores, it dries out, and in its place grows a summer stem from the rhizome of the plant. With the help of a magnifying glass, summer and spring stems and lateral branches are observed, their similarities and differences are identified. Spring and summer stems, structure, differences, shape and location of leaves are drawn.

The microscopic slide is prepared from a cross-section of a steamed or freshly picked stem and examined first under a small lens and then a large lens of microscope. Attention is paid to the air space in the center of the stem. On the outside of the stem in the central direction is the location of the tissue - the epidermis, photosynthetic-assimilating tissue, along with the air space, the parenchymal tissue of the cortex. Attention is paid to the order of the phloem and xylem in the **vascular bundle**, its type, whether the bundle is open or closed. The cross-section structure of the stem seen under a microscope is drawn on an album sheet.



Figure 6. General view of *Equisetum arvense*

In herbarium and ready micropreparations, the longitudinal section of the sporous capsule of *E. arvense* is studied using a magnifying glass and a microscope. Attention is paid to the shape of the sporophylls and the order of their location on the axis of the spike. A single sporophyll is isolated, and its polygonal thyroid shape, with up to eight sporangia inside, is seen and counted. A sporangious sporophyll with its seta is drawn in the album.

The sporangium wall is three-layered and initially contains archeospore cells. The inner layer of the wall becomes slimy and single-layered. Spore-forming cells (sporogenes) are divided by reduction to form haploid spores.

The sporophyll of the *E. arvense* is taken out and placed on a glass slide, and ruptured with a drug needle. Spores taken out of the sporangium can be seen through a small and a large microscope lens. The shape, size, color, three-layered membrane of spores are determined. The outer layer of the spore separates to form two ribbon-elaters. The object is lightly blown on the glass, dried, and spores are

observed. The structure of the spore, and the elaters are drawn. Attention is paid to the uniformity of the spores.

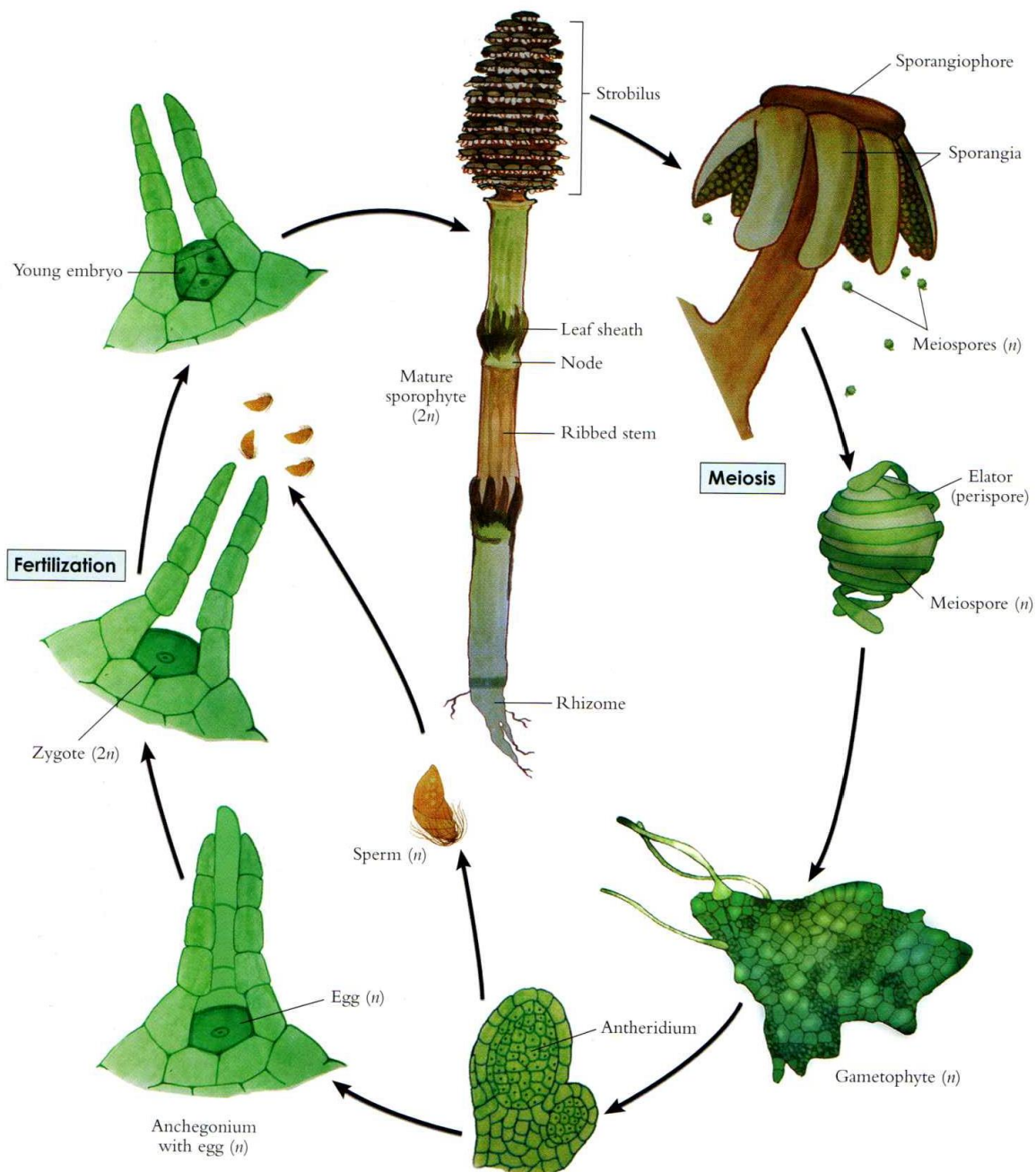


Figure 7. Structure and reproduction of *Equisetum arvense*

GAMETOPHYTE. The gametophytes of the *E. arvense* are also sexual generation formed of spores. It has a haploid set of chromosomes. To The study the gametophytes (sexual reproduction) prepared microscope slides, drawings in books and sketches will be used.

Although the spores are the same, there are two types of gametophytes - antheridic and archegonic gametophytes. Hence, spores have 2 different physiological characteristics. Whether gametophyte is monoecious or dioecious, has presence or absence of rhizoids, has the ability to photosynthesize, the similarities and differences of antheridic and archegonic gametophytes are identified.

The egg cell in the archegonium and the sperm formed in the antheridia are observed. Gametophytes, the structure of sexual organs and sexual cells is drawn on an album sheet.

When fertilization occurs, the zygote first forms a young sporophyte and then a mature photosynthesizing sporophyte.

Self Testing Questions:

1. What is the structure of summer and spring stems of *Equisetum arvense*?
2. How does reproduction occur in *E. arvense*?
3. What do gametophyte and sporophyte of *E. arvense* consist of?

DIVISION *POLYPODIOPHYTA* (*PTERIDOPHYTA*) - Ferns

Among the divisions of higher plants, species diversity of ferns stands in third place after magnoliophytes (flowering plants) and bryophytes (mosses). Currently, there are about 300 genera and more than 10,000 species of ferns on Earth. They are also the oldest plants. This is because representatives of the ferns, which have not survived, have grown and formed forests at the same time as the ancient horsetails, which became extinct during the Devonian period (415-370 million years ago). Ferns are found in mountains, plains, swamps, and watery areas, especially they are widely distributed in forests of tropical and subtropical climatic zones.

As ferns grow in a diverse ecological environments, different life forms can be found among their plant species. Especially in tropical and subtropical conditions, from perennial herbaceous species with standing and horizontal stems to epiphytes and lianas, several genera from *Cyataceae* family with woody fern representatives going up to 25 meters in height.

In Central Asia, only herbaceous plants are represented. As in horsetail plants, the root system of ferns is well developed. Representatives of some plants from *Ophioglossaceae* and some ancient primitive groups have thick bark and roots.

One of their characteristic features is that the leaves are large and the growing point is located at the top of the stem. In this respect they are different from *Lycopodiophyta* and *Equisetiophyta*. They are also phylogenetically related to rhinophytes due to changes in the leaf thallus. The leaves of some ferns are up to 30 m long. In most ferns, the leaf performs two functions, namely photosynthesis and spore-forming, while in others the spore-forming leaves lose chlorophyll and perform only asexual reproduction (e.g. in *salvinia*).

As in *Lycopodiophyta* and *Equisetiophyta* divisions, sporophyte generation is predominant in the plant species of *Polypodiophyta* division. During evolutionary development, the sporangia of ferns have moved from the edge of the leaf to the bottom. The location of the

sporangia on the underside of the leaf makes it easier to protect them from the adverse effects of the external environment, while the latter allows for the normal course of photosynthesis on the surface of the leaf. Sporangia in clusters on the leaves are called *sorus* (Greek *soros* - a bunch).

In ferns, the sperm are as numerous as in horsetails, and fertilization takes place with the help of moisture. Several egg cells of the gametophyte can be fertilized, but only a single zygote will be developed and forms prothallus or gametophyte. The prothallus settles in the ovary, from which a sporophyte develops.

***Polypodiophyta* division consisted of 7 classes:**

1. *Aneurophytopsida*
2. *Archaeopteridopsida*
3. *Clodoxylopsida*
4. *Zigopteriopsida*
5. *Ophioglossopsida*
6. *Marattiopsida*
7. *Polypodiopsida*

Representatives of 1-4 classes from the above are completely extinct. Representatives of the class *Aneurophytopsida* are the oldest ferns. They grew in the middle of the Devonian period (400-375 million years ago). Representatives of the *Archaeopteridopsida* class are also extinct. They were distributed from the end of the Middle Devonian to the beginning of the Carboniferous (320-360 million years ago). Representatives of this class were similar in appearance to modern conifers.

Representatives of the class *Clodoxylopsida* also grew almost at the same period with the classes of *Aneurophytopsida* and *Archaeopteridopsida*.

Representatives of the ferns of *Zigopteriopsida* class existed from the end of the Devonian period to the Permian.

6-CLASS

TOPIC: STRUCTURE AND DEVELOPMENT OF *DRYOPTERIS FILIX-MAS*

Division: *Polypodiophyta* (*Pteridophyta*)

Class: *Polypodiopsida*

Subclass: *Polypodiidae*

Order: *Polypodiales*

Family: *Polypodiaceae*

Species: *Dryopteris filix-mas*

Purpose of the class: to study the structure, development stages and classification of *Dryopteris filix-mas*.

Objectives of the class: to teach students herbarium specimens, and to inform them about live materials and classification of the target plant; to strengthen the student's knowledge of the subject; to be able to answer the teacher's questions.

Demonstration materials and equipments: powerpoint slides; animations, online materials. Sketch of reproduction cycle of *D. filix-mas*, herbarium specimens, plant exhibits, prepared microscope slides (antheridium, archegonium, sporous capsule), microscopes, magnifying glass, microscope slides, cover slips, preparation needles, water dropper (pipette).

LEARNING PROCEDURE

Students will get to know the herbarium specimens and exhibits of *Dryopteris filix-mas* and learn about the biological characteristics of gametophyte and sporophyte using the alive plants, teaching materials and books.

SPOROPHYTE: The sporophyte generation of *D. filix-mas* is a perennial herb with a complex set of serrated leaves (Fig. 8). The leaves protrude from the tip of the rhizome. The rhizome (deformed stem) is located horizontally close to the surface, is dark brown and forms thin

additional roots. The rhizome is densely covered by the base of the leaf blades left over from previous years. As it can be seen in the picture, the young leaves that grow from the rhizome are twisted like a worm in the first year, covered with thick brown coin-shaped skins and elongated from the base to the tip. The mature leaves are compound pinnate, 1,0-1,2 m long. Microscopic slide prepared from **the steamed rootstock** of *D. filix-mas* or the ready-made microprepared one is viewed under a microscope, first through a small, then through a large lens. The rhizome has a much more complex internal structure. The surface is covered with epidermis, the cortex is divided into outer and inner parts.

As can be seen under a microscope, the set of cells on the outside of the cortex has a function of mechanical tissue. There are many vascular bundles of different sizes. The larger ones are located in the center of the stem, while the smaller ones are scattered along the rhizome. In the microscopic slide can be seen that the vascular bundle is bordered with endoderm, and next to endoderm there is the pericycle consisting of single or two layer cells, which is further followed by the phloem-sieve tubes and lub parenchyma. The xylem is located in the center of the vascular bundle. The structure of the cross-section of the rhizome, seen under a microscope, is drawn on an album sheet.

When we look at the anatomical structure of a *D. filix-mas* leaf under a microscope, we see that its upper and lower sides are covered with epidermis, and that their cells are rich in chlorophyll granules. The mesophyll has much larger intercellular spaces. At the base of the leaf, around the central veins, are soruses (a collection of sporangia). The soruses are round, heart-shaped, and covered with a thin colorless membrane (indusium). The archesporial cell in the sporangia divides in succession to form 16 **parent cells**. They, in turn, are reduced to form tetra-spores. The spore is a single-celled structure surrounded by two shells: the inner endospore and the outer exosporium.

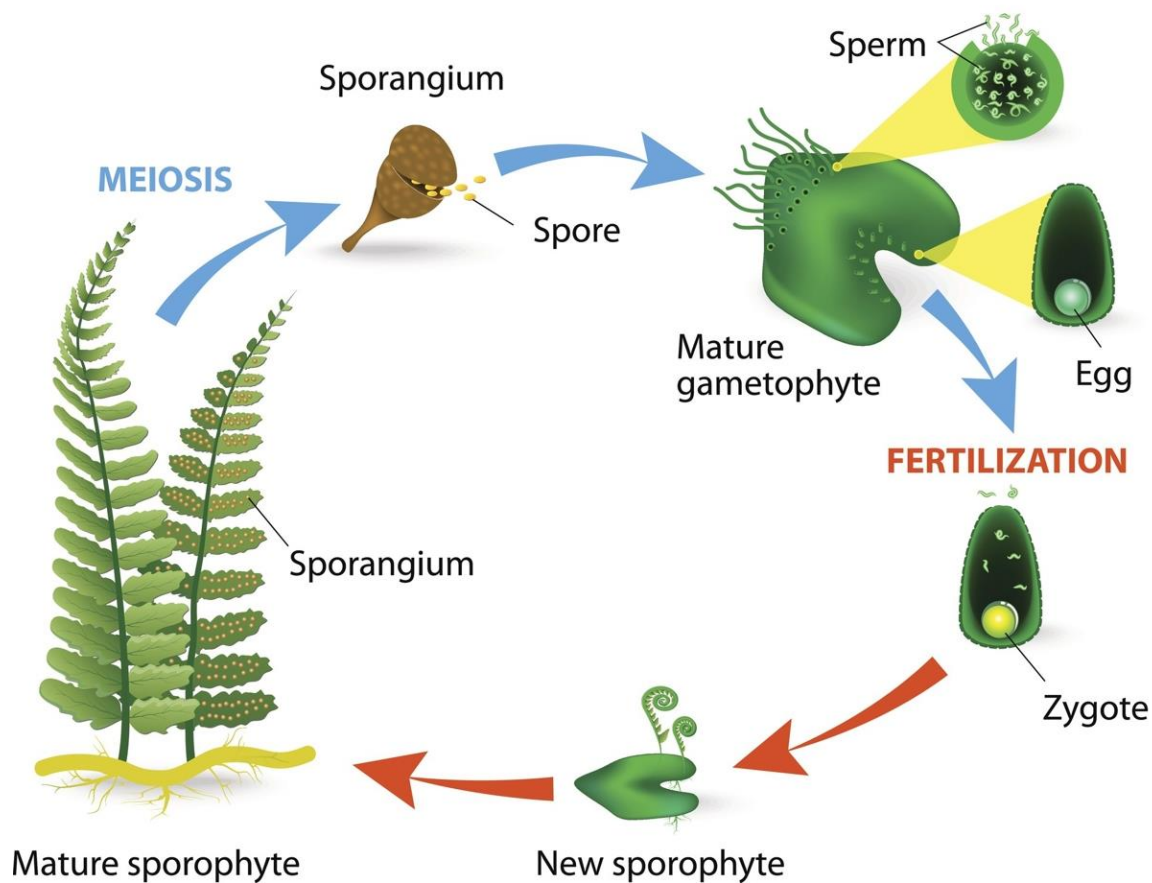


Figure 8. Development cycle of *Dryopteris filix-mas*

GAMETOPHYTE: The spores spill on the ground, germinate under conditions of sufficient moisture and heat, and form a sexual prothallus - gametophyte. Gametophyte is green, rhizoid, round-hearted, in the form of a plate with a diameter of 0,5-0,9 cm, single-celled, and is able to photosynthesize.

Archegonium and antheridium get matured in it. The prothallus is adapted to independent living, consisting of single-layered cells at the edge and multilayered cells in the middle. The archegonia is **flask-shaped**, with the **ventricle** embedded in the prothallus tissue. In the archegonia, an egg cell is formed, and in the antheridium, many monoflagellate spermatozoon develop. The **neck cells** in the archegonia melt and form a fluid that holds the sperm. When the antheridium ruptures, the sperm is released into the water, and along the archetypal neck, the egg moves toward the cell and fertilizes it. As a result, the zygote is formed and the sporophyte begins to reproduce. Zygote cells

divide by mitosis to form a young sporophyte and then to an adult sporophyte plant. In it, the set of chromosomes is always diploid. *D. filix-mas* also reproduces vegetatively using the rootstock. Diagram of the view and development of the *D. filix-mas* needs to be drawn.

Self Testing Questions:

1. What does the sporophyte of *Dryopteris filix-mas* consist of?
2. What does the gametophyte of *D. filix-mas* consist of?
3. Explain the structure and development stages of *D. filix-mas*.
4. Explain the anatomical and morphological leaf structure of *D. filix-mas*.

DIVISION *PINOPHYTA* (*GYMNOSPERMATOPHYTA*) - *Exposed Seed Plants*

Pinophyta is the oldest group of seed plants, their fossils have been found in the Devonian period of the Paleozoic era. The microfilic and macrophilic line of evolution continues inside the division. These are Pine-like descended from Pteridophyta and are in their organization an intermediate stage between ferns and angiosperms. Pine-like are the most complex archegonal plants in structure. Among them there are trees, partially shrubs growing on land, but there are none growing in water.

The sporophyte is well developed and predominant above the gametophyte. This division includes plants with a root, stem, and leaf. The gametophyte is reduced and depends on the sporophyte. One of the most important features is that, in representatives of this division, a seed is formed, from the embryo from which a sporophyte develops. The seed is a new formation, and it was absent from all the higher plants studied so far. The seed is a developed metamorphosed megasporangium. The body of pine trees is always woody, so they live long. They also have a very complex anatomical structure. In the species living today, there are eustel and collateral vascular bundles. The activity of the cambium has led to thickening and secondary tissues, the formation of xylem and floma, and the formation of the periderm. The wood is composed of tracheids. There are no true **vessels** or vascular bundles (trachea) in the *Pinophyta*.

The root system consists of the main and lateral roots. In terms of anatomical structure, they are similar to flowering plants. In species that grow in the tropics, the leaves are pinnate, feathery, growing from the top, like in ferns. In the anatomical structure of the leaf there is a folded parenchyma and complex vascular bundles. *Pinophyta* plants do not have vegetative reproduction.

There are 6 classes in *Pinophyta* division and it includes up to 800 species. Of these, representatives of 2 classes are found only in the fossil form.

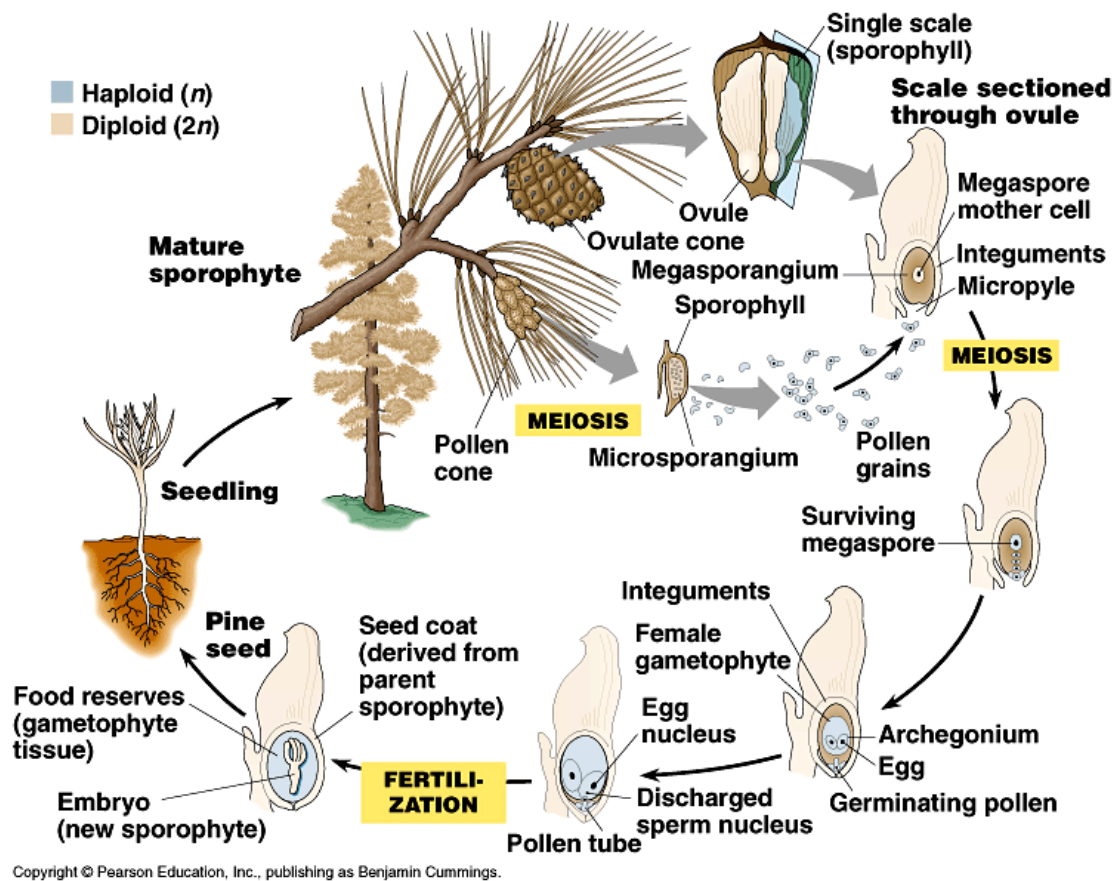


Figure 9. Development cycle of gymnosperms

The division is classified as follows:

1-Class: Pteridospermatopsida

Orders: a) Pteridospermales

b) Caytoniales

2-Class: Cycadopsida

Order: Cycadales

3-Class: Bennettitopsida

Order: Bennettitales

4-Class: Ginkgopsida

Order: Ginkgoales

5-Class: Pinopsida

Sub class: Cordaitidae

Order: Cordaitales

Under the class Pinoideae
(Coniferoideae)

a) Order: Pinales (Coniferales)

6-Class: Gnetopsida

a) Order: Ephedrales

b) Order: Gnetales

c) Order: Welwitschiales

7-CLASS

TOPIC: STRUCTURE AND DEVELOPMENT OF *GINKGO BILOBA*

Division: *Pinophyta (Gymnospermartophyta)*

Class: *Ginkgopsida*

Order: *Ginkgoales*

Family: *Ginkgoaceae*

Genus: *Ginkgo*

Species: *Ginkgo biloba*

Purpose of the class: to study the structure, development stages and classification of *Ginkgo biloba*.

Objectives of the class: to teach students herbarium specimens, and to inform them about live materials and classification of the target plant; to strengthen the student's knowledge of the subject; to be able to answer the teacher's questions.

Demonstration materials and equipments: powerpoint slides; animations, online materials. Sketch of reproduction cycle of *G. biloba*, herbarium specimens, plant exhibits, prepared microscope slides (antheridium, archegonium, sporous capsule), microscopes, magnifying glass, microscope slides, cover slips, preparation needles, water dropper (pipette), diagram of its development cycle.

LEARNING PROCEDURE

Students will get to know the herbarium specimens and exhibits of *Ginkgo biloba* and learn about their biological characteristics using the teaching materials and books.

SPOROPHYTE: Get acquainted with the exhibits and the *G. biloba* tree or herbarium specimen from a textbook, study its structure and reproduction. Ginkgo is a tree growing upto 30 - 40 m, with a smooth dark gray bark, has monopodial branching. The anatomical structure of the ginkgo stem is as follows: it is similar to that of conifers:

the bark and pith are poorly developed, and the secondary xylem is more powerful. When considering the herbarium, attention is drawn to the fact that its shoots are of two genera: a) elongated, with leaves scattered on them, b) shortened, with leaves in bunches.

The leaves have long petioles, triangular in outline, bilobate at the apex, with a very ancient fan-shaped dichotomous venation. Leaves falling for the winter, which is quite a rare occurrence in gymnosperms.

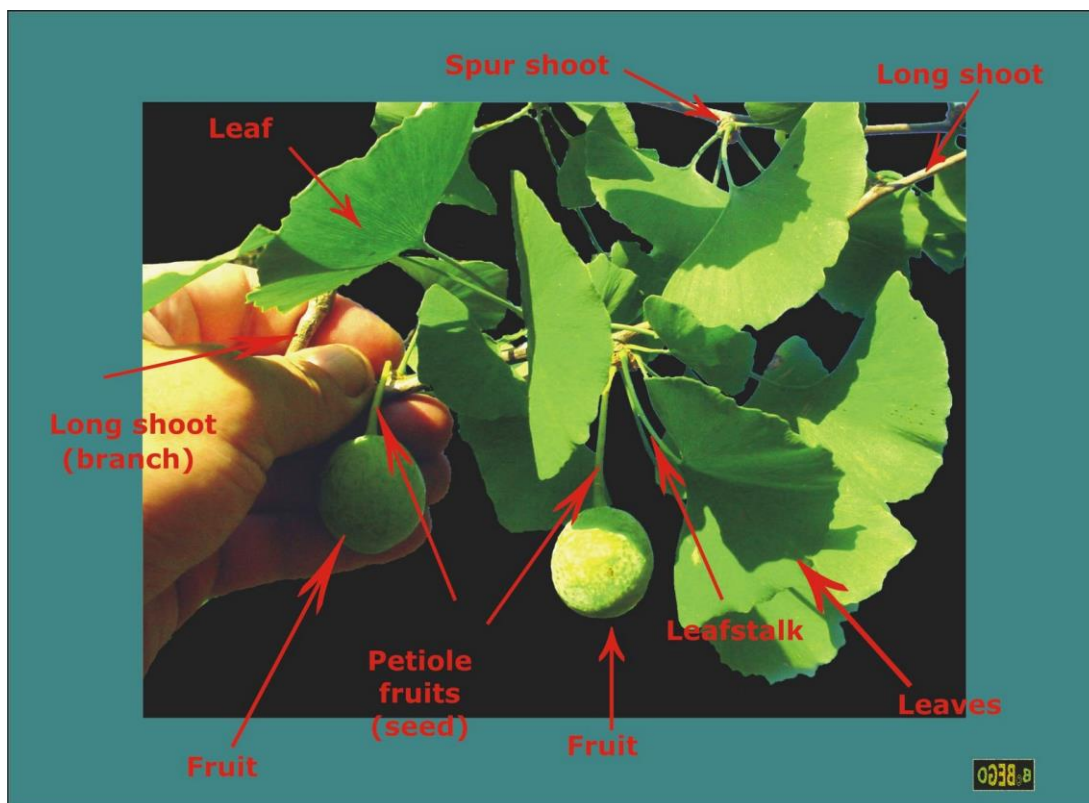


Figure 10. Morphological parts of *Ginkgo biloba*

Ginkgo is a dioecious plant. Microsporophylls are collected in a group, like the inflorescence of an earring, megasporophylls - on a long stem, usually with two megasporangia (ovules). Microsporangia are located in microsporophylls. The ovules have a typical structure and are similar to the ovules of a cycad.

As can be seen from the figure, the microspore or pollen grows already in the microsporangia and divides to form 3 cells: protallial, anteridial and vegetative. In this form, pollen spreads and in various ways can enter the pollen duct of the ovule.

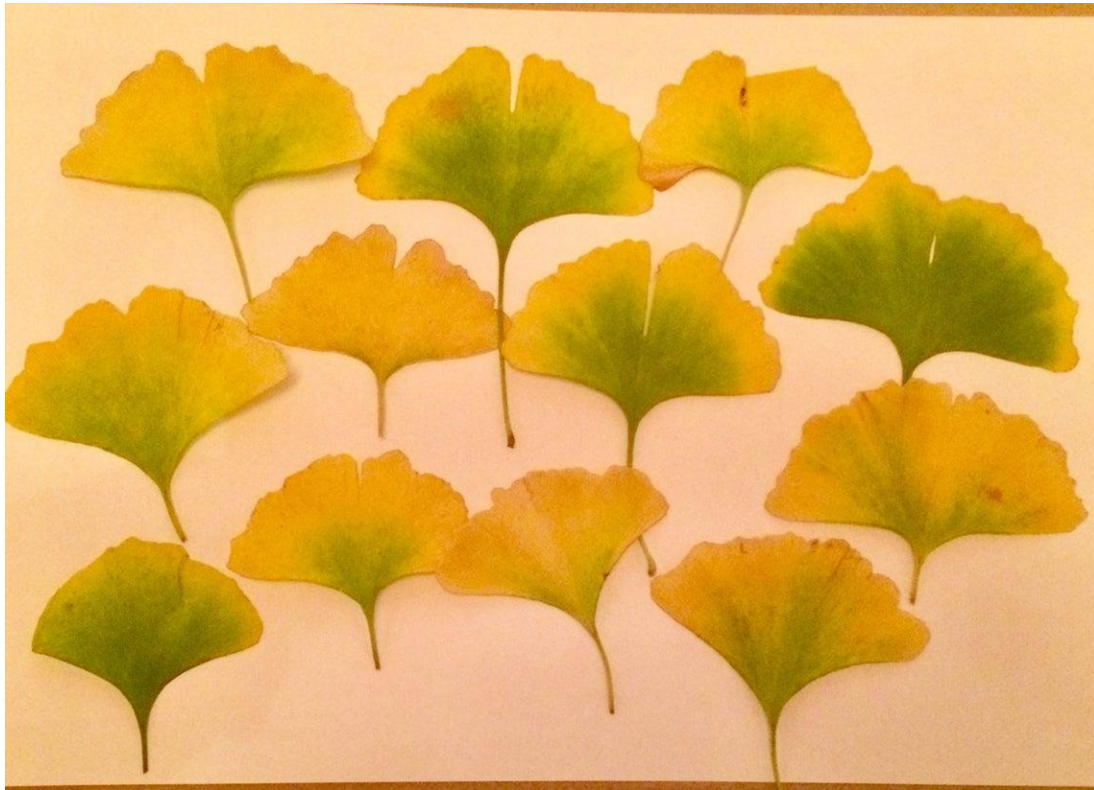


Figure 11. Different leaf shapes of *Ginkgo biloba*

GAMETOPHYTE: In the ovule, pollen (microspore) grows and forms a male gametophyte. The ovule is surrounded by a dense shell - the integument; it is not closed at the apex and a micropyle (pollen duct) is formed there. Nucellus is located inside the integument. The upper cells of nucellus, on dissolving, form a pollen chamber, and from the lower ones, the mother cell of megasporangia is formed. At first it forms 4 megaspores, then 3 of them dissolve. One cell continues to divide and forms the primary endosperm, in the upper part of which two archegonia develop.



Figure 12. Female and male flowers of *Ginkgo biloba*

The microspore trapped in the pollen chamber continues to grow. From the anteridial cell, two multi-flagellate spermatozoon are formed. They are located in the plasma of the haustorium. The tip of the pollen chamber breaks open and the entire contents are poured onto the endosperm. One of the sperm will fertilize the egg of one of the archegonia - a zygote is formed. The second archegonium and sperm dissolve. In the process of dividing the zygote, an embryo is formed. The newly formed seed has an outer and inner layer that is soft, juicy, aromatic, and the middle layer is stony. In appearance, the seed looks like a plum, in China and Japan, the soft part of it is eaten. Inside the seed is an embryo with two cotyledons and a primary endosperm that feeds the embryo. The cotyledons remain in the ground during germination.



Figure 13. Male inflorescence of Ginkgo biloba

Sketch the structure and reproduction cycle of *G. biloba* in an album (Fig. 14).

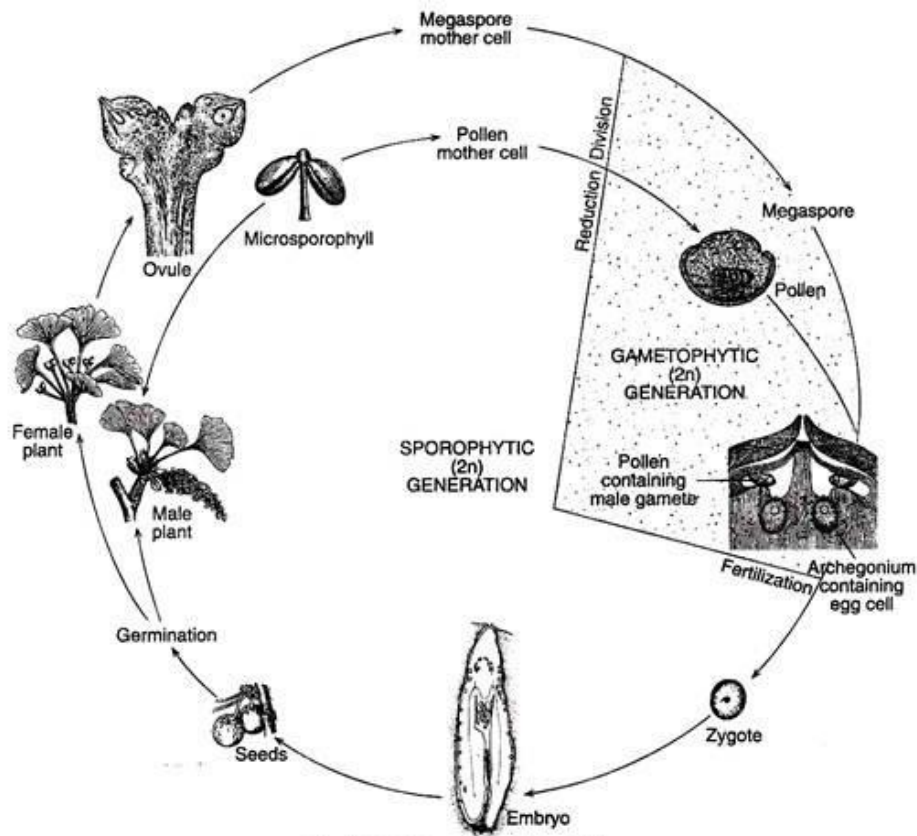


Figure 14. The structure and reproduction of *Ginkgo biloba*

Self Testing Questions

1. Describe the structure of *Ginkgo biloba*.
2. What is the development cycle of *G. biloba*?
3. Explain the sporophyte structure of *G. biloba*.
4. What are the specific features of the stems and leaves of the plant?
5. Describe the structure of microsporophyll and ovule of *G. biloba*.

8-CLASS

TOPIC: STRUCTURE AND DEVELOPMENT OF *PINUS SILVESTRIS*

Division: *Pinophyta* (*Gymnospermatophyta*)

Class: *Pinopsida* (*Conifers*)

Order: *Pinales (Coniferalis)*

Family: *Pinaceae*

Genus: *Pinus*

Species: *Pinus silvestris*

Purpose of the class: to study the structure, development stages and classification of *Pinus silvestris*.

Objectives of the class: to teach students herbarium specimens, and to inform them about live materials and classification of the target plant; to strengthen the student's knowledge of the subject; to be able to answer the teacher's questions.

Demonstration materials and equipments: powerpoint slides; animations, online materials. Sketch of the reproduction cycle of *P. silvestris*, herbarium specimens, plant exhibits, prepared microscope slides (antheridium, archegonium, sporogonium), microscopes, magnifying glass, microscope slides, cover slips, preparation needles, water dropper (pipette), diagram of its development cycle.

LEARNING PROCEDURE

Students will get to know the herbarium specimens and exhibits of *Pinus silvestris* and learn about their biological characteristics using the teaching materials and books.

SPOROPHYTE: Forest pine is a tree reaching 20 - 40 m in height and 1 - 2 m in diameter, has monopodial branching. Has shoots of two genera: shortened and elongated. Anatomically, the stem of the common pine is characterized by a small development of the bark and pith, but a large secondary xylem deposited by a well-developed cambium. The wood consists of tracheids, the primary wood is annular and spiral, the secondary wood has bordered pores on the radial walls; there are no vessels. The wood parenchyma is poorly developed; in some it is completely absent. The heart-shaped rays are well developed. Phloem consists of sieve tubes without companion cells. Pine has clearly pronounced annual layers, especially in those growing in conditions of changing seasons. Also, pine is characterized by resin passages that arise

schizogenically, as well as the formation of resins and essential oils. Resin passages are frequent in bark, wood, and needles. The root of the forest pine - main with lateral, and Mycorrhiza is developed. The anatomical structure of the roots is the same as that of flowering plants. Pine leaves are called needles, they are flat, tetrahedral, on shortened shoots which have two, or sometimes 3 leaves. When examining the cross section of the leaf, we see the following: they have a peel with stomata outside, behind it is a layer of mechanical cells (hypodermis) and resin ducts, then a mesophyll of folded chlorenchyma with sinuous outlines of cell walls that form outgrowths inside the cells ~~is located~~, in the center there is a conductive bunch. Outside, it is covered with endoderm. The bundle is collateral and double - it consists of two clusters of xylem, a narrow strip of cambium and bast, and is surrounded by the entire sclerenchyma. Pine leaves have a xerophilic structure: hard, leathery, perennial 3-6 years and fall off alternately. Forest pine reproduces only by seeds, as in other pine-trees, sporophytes dominate in the reproduction cycle. The highly reduced gametophyte lives off the sporophyte. Pine is a monoecious dioecious plant, on the shoot either cones with microsporophylls or with megasporophylls are always formed. Cones with microsporophylls are yellow or reddish in color. On the underside of the microsporophyll there are 2 microsporangia each and microspores (pollen) are formed in its nests. The male gametophyte begins to develop already in the microsporangia. The microspore has 2 shells - the outer one is exine, the inner one is intine. By the divergence of these shells from one another, air sacs are formed - increasing the surface of the pollen, they contribute to its transfer by the wind over long distances. When microspores germinate, pollen is formed, during the division of which, protallial rhizoid cells are formed, then they dissolve. During the second division of the pollen nucleus, vegetative cells of the outgrowth are formed, then from other cells of the anteridial pollen, which in turn is divided into two sperm without flagella, and due to the vegetative cell, the pollen tube grows further. In this form, the pollen is distributed.

GAMETOPHYTE: The cone consists of megasporophylls, which are located at the top of the shoot. Cones are a longitudinal axis on which in the axils of the covering scales, shoots sit - seed scales, which then grow and become woody. On the upper side of the seed scales, there are 2 megasporangia, i.e. ovules.

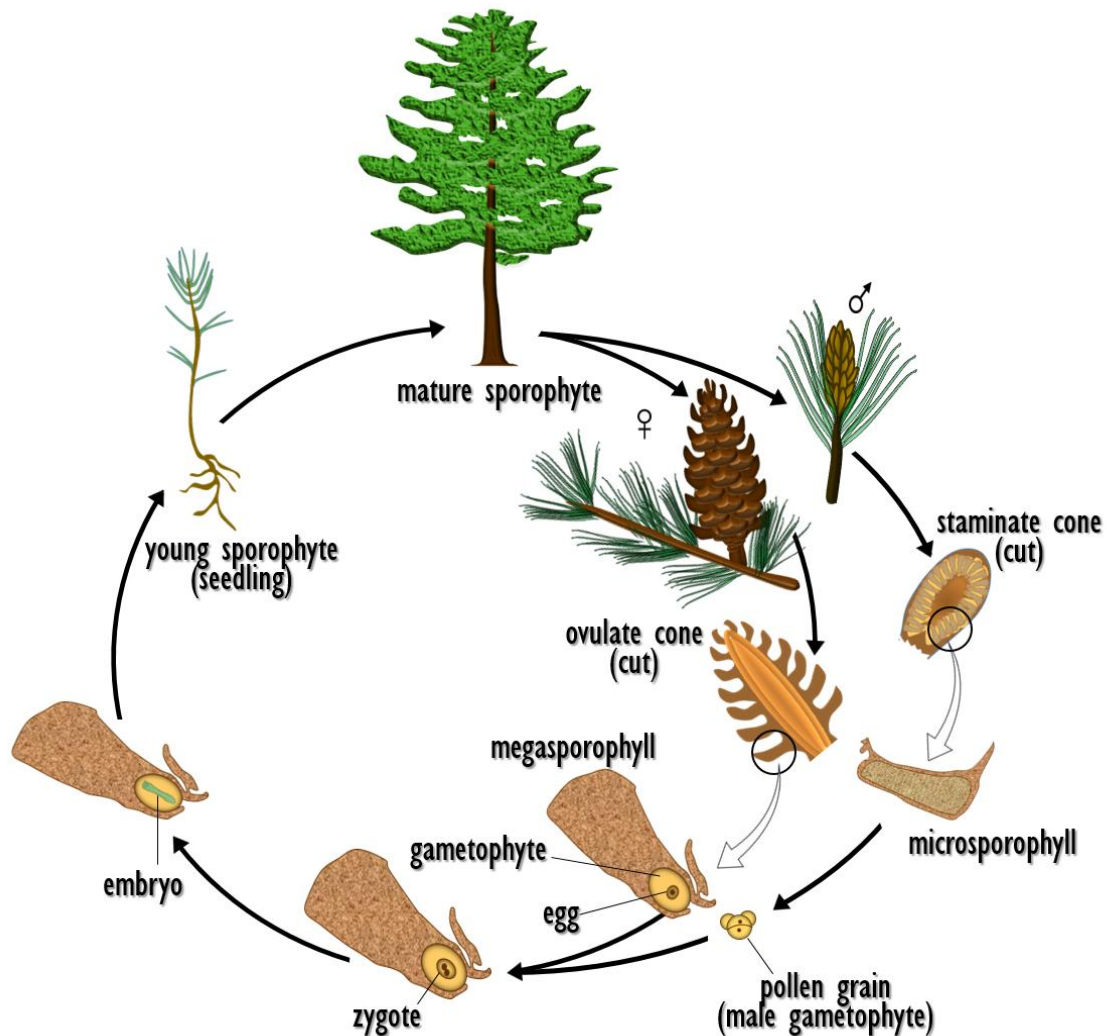


Figure 15. Development cycle of *Pinus silvestris*

The ovule is covered with integument, which consists of 3 layers: medium hard and dense, the other two are softer, sometimes juicy. The pollen chamber under the micropyle is poorly developed. In nucellus, a large megaspore develops and a female gametophyte is formed - the primary endosperm, and in it two archegonia, very reduced, with poorly developed tubular cells. Primary endosperm well developed goes to feed the developing embryo. The pollen, hitting the ovule by the wind, is

drawn into the pollen chamber by the liquid secreted by the nucellus. Here, the pollen grows into the pollen tube, which serves to carry the flagellate sperm to the egg. When the pollen tube grows to the archegonium, one of the sperm leaves it and merges with the egg, and the second sperm and archegonia die. A long period occurs between pollination and fertilization in the forest pine. The ovules reach full development only in the second year, and the pollen trapped in it stops growing. Fertilization occurs after winter dormancy, and male and female gametophytes continue to develop 13 to 15 months after pollination. But the fertilized egg immediately divides and forms an embryo with all its parts (the embryonic main root, stalk, hypocotyl knee, kidney and from 2 to 18 cotyledons, as well as a developed suspension, which presses the embryo into the endosperm, which goes to feed the embryo. The ovule turns into a seed, and the cover gives the skin of the seed, often woody or leathery appearance. Sketch in the album the structure and breeding cycle of the forest pine (Fig. 15).

Self Testing Questions:

1. Describe the structure of *Pinus silvestris* sporophytes and gametophytes.
2. What is the structure of the branches and leaves of *P. silvestris*?
3. Explain the development cycle of *P. silvestris*.
4. What is the structure of the *P. silvestris* sporophyte?
5. How is the gametophyte of a *P. silvestris* formed?
6. What is the structure of the megasporophyll of *P. silvestris*?

DIVISION MAGNOLIOPHYTA (ANGIOSPERMATOPHYTA) – ***Flowering plants***

The plants belonging to this division are also called the Angiosperms or Flowering Plants. This group is the most diverse, ecologically dominant and economically important of all the living plant division. The designation of Magnoliophyta comes from the most ancestral family in this taxon. The flowering plants are divided into two class: Magnoliopsida (dicots) and Liliopsida (monocots).

Magnoliophyta are the second group of seed plants and are widespread throughout the world due to the complex structure of the body, the specificity of sexual reproduction (the presence of a flower), the number of species and the great potential for adaptation to environmental conditions. These differ sharply from higher spores and pines. Phylogenetically they are the youngest of all higher plants, and their rapid development dates back to the second half of the Cretaceous. At the beginning of the XX century the number of species of flowering plants was estimated at 160 thousand (N. Kuznetsov), by the middle of the century there were more than 200 thousand and today there are 533 families, and about 250 000 species belonging to 13 000 genera (A.L.Takhtadjan, 1987).

Another important feature of magnolias that distinguishes them from other higher plants is the presence of a flower that has a special sexual reproductive organ. As with all higher plants, there are two phases in their development cycle: the sporophyte (diploid, asexual) and the gametophyte (haploid, sexual). These two phases (phases) alternate and galvanize independently of the droplet aqueous medium.

The flower is a combination of metamorphosed stems and leaves and is a generative branch that develops from a bud. The following are some of the features that make them different from other higher plants:

1) ovule is one or more, and is located inside the ovary; 2) pollen grains fall and settle directly on the stigma of pistil, but not on the micropile of ovule; 3) gametophytes are very short, without gametangia, and the gametophyte is formed as a result of several mitotic divisions; 4)

double fertilization occurs, one of the sperm joins with the egg cell inside the ovarian sac and the other with the central cell of the diploid chromosome, resulting in the formation of the ovary and the triploid endosperm tissue that feeds it.

The flowers of flowering plants are the most remarkable feature distinguishing them from other seed plants. Flowers aided angiosperms by enabling a wider range of evolutionary relationships and broadening the ecological niches open to them, and allowing flowering plants to eventually dominate terrestrial ecosystems.

Stamens are much lighter than the corresponding organs of gymnosperms and have contributed to the diversification of angiosperms through time with adaptations to specialized pollination syndromes, such as particular pollinators. Stamens have also become modified through time to prevent self-fertilization, which has permitted further diversification, allowing angiosperms to eventually fill more niches.

The male gametophyte in angiosperms is significantly reduced in size compared to those of gymnosperm seed plants. The smaller pollen decreases the time for pollination – the pollen grain reaching the female plant – to fertilization of the ovary; in gymnosperms fertilization can take up to a year after pollination, while in angiosperms the fertilization begins very soon after pollination. The shorter time leads to angiosperm plants setting seeds sooner and faster than gymnosperms, which is a distinct evolutionary advantage.

The closed carpel of angiosperms also allows adaptations to specialized pollination syndromes and controls to prevent self-fertilization, thereby maintaining increased diversity. Once the ovary is fertilized, the carpel and some surrounding tissues develop into a fruit, and this provides another opportunity for angiosperms to increase their domination of the terrestrial ecosystem with evolutionary adaptations to dispersal mechanisms.

The reduced female gametophyte, like the reduced male gametophyte may be adaptations allowing for more rapid seed set, eventually leading to such flowering plant adaptations as annual

herbaceous life cycles, allowing the flowering plants to fill even more niches.

The flowers are unisexual, bisexual, sometimes asexual; they can be arranged in simple or complex inflorescences. The structure of the flower, the number of parts and the arrangement are indicated by the flower formula and diagram. For example, the formula of apple flower is: **$\text{Ca}_5\text{Co}_5\text{A}_\infty\text{G}_\infty$** .

Flowers show remarkable variation in form and elaboration, and provide the most trustworthy external characteristics for establishing relationships among angiosperm species. The function of the flower is to ensure fertilization of the ovule and development of fruit containing seeds. The floral apparatus may arise terminally on a shoot or from the axil of a leaf. Occasionally, as in violets, a flower arises singly in the axil of an ordinary foliage-leaf. More typically, the flower-bearing portion of the plant is sharply distinguished from the foliage-bearing or vegetative portion, and forms a more or less elaborate branch-system called an inflorescence.

Double fertilization refers to a process in which two sperm cells fertilize two cells in the ovary. The pollen grain adheres to the stigma of the carpel (female reproductive structure) and grows a pollen tube that penetrates the ovum through a tiny pore called a micropyle. Two sperm cells are released into the ovary through this tube. One of the two sperm cells fertilizes the egg cell, forming a diploid zygote or embryo, also called the ovule. The other sperm cell fuses with two haploid polar nuclei in the center of the embryo sac. The resulting cell is triploid ($3n$). This triploid cell divides through mitosis and forms the endosperm, a nutrient-rich tissue inside the fruit. When seed develops without fertilization, the process is known as apomixis.

As the development of embryo and endosperm proceeds within the embryo-sac, the sac wall enlarges and combines with the nucellus (which is likewise enlarging) and the integument to form the seed-coat. The ovary wall develops to form the fruit or pericarp, whose form is closely associated with the manner of distribution of the seed.

Frequently the influence of fertilization is felt beyond the ovary, and other parts of the flower take part in the formation of the fruit, e.g. the floral receptacle in the apple, strawberry and others.

The character of the seed-coat bears a definite relation to that of the fruit. They protect the embryo and aid in dissemination; they may also directly promote germination. Among plants with indehiscent fruits, the fruit generally provides protection for of the embryo and secures dissemination. In this case, the seed-coat is only slightly developed. If the fruit is dehiscent and the seed is exposed, the seed-coat is generally well developed, and must discharge the functions otherwise executed by the fruit.

Classification of Magnoliophyta

The classification of *Magnoliophyta* (flowering plants) and the establishment of their phylogenetic relationships depend on the phylogenetic significance of individual systematic characters. These traits can be primitive and ancient, or relatively young in origin. But the views of different scientists on this are very different and it is sometimes very difficult to establish appropriate connections between individual systematic groups on this basis. Only in A.Engler the phylogenetic system, which is now considered obsolete, has been reduced to individual genera and sometimes species. The largest herbariums in many countries, as well as the herbarium of the Scientific and Production Center “Botanika”, are based on the Engler system and include 1 million copies of the herbarium.

However, it is possible to observe primitive or complex, oldest and youngest, in origin characters sometimes within one taxonomic group or taxon, or even in one species. For example, in many *Ranunculaceae* flowers are **actinomorphic**, but in *Delphinium* and *Aconitum*, on the contrary, they are zygomorphic; in some representatives of *Nymphaeaceae* the ovary is superior, in *Victoria regia* the ovary is inferior. From *Rosaceae* family the genera of *Fragaria*, *Rubus* have superior ovary, in the genera of *Malus*, *Pyrus*, *Rosa* the ovary is inferior.

In most species of Boraginaceae the flowers are actinomorphic, but in the genera of *Lycopsis* and *Echium* the flowers are zigomorphic.

Among the modern phylogenetic systems of flowering plants, the system of the Russian scientist Academician A.L. Takhtadzhyan is distinguished by its perfection and scientific validity.

Most taxonomists divide Magnoliophyta division (flowering plants) into two classes:

1. Magnoliopsida or Dicotyledons
2. Liliopsida or Monocotyledons

Based on the latest advances in the taxonomy of flowering plants between Dicotyledons and Monocotyledons, there are the following differences and exceptions in traits:

1. The seed of the Magnoliopsida has two cotyledons, the Liliopsida has single cotyledon. But sometimes some Magnoliopsida have a monocotyledonous seed. For example, this occurs in *Ficaria* from *Ranunculaceae*, in *Corydalis* from *Fumaraceae*, in *Cyclamen* from *Primulaceae* and in general, in the example of 40 plants, this structure can be observed. In Liliopsida, presence of dicotyledonous seeds was found in *Agaphanthus* and *Cythranthus*. Despite the exceptions, this feature is the most important distinguishing feature for both the groups.

2. In Dicotyledonous plants, the cotyledons are lateral (have lateral placement), in Monocotyledons, they are terminal (located at the top of the seed).

3. In Magnoliopsida, the stem and root have a secondary thickening; In Liliaceae, the main root dies off and no cambium activity is observed. The adventitious roots form a fibrous root system. As an exception, thickening occurs in treelike lily dracaena (*Dracaena*) and cordilina (*Cordylina*) by neoplasm of vascular fibrous bundles in the peripheral part of the stem. But in 23 species of Monocotyledonous plants, some Lily plants, lilies (*Lilium*), corn (*Zea mays*), cattail (*Typha*) and others, the cambium exists, but soon ceases to function. Representatives of this family are Buttercup (*Ranunculaceae*) and podophyllum (*Podophillum*) with no cambium.

4. Leaves of *Magnoliaceae* have a varied structure: simple, complex, petiolar, sessile, has a sheath, stipules, whole, dissected, and separate; The venation is parallel, arcuate, pinnate, and palmate. Lily-shaped leaves are often vaginal, without stipules, simple, entire-edged, with parallel or arcuate venation, and dissected leaves are found only in Palms and Aroids.

However, in dicotyledonous leaves, they can be vaginal, without stipules, narrow, parallel or arcuate venation, for example, the representatives of this family are *Caryophyllaceae* and *Plantaginaceae*.

5. Flowers of *Magnoliaceae* are five-circular or four-circular, five, four-membered (when the flower is circular, i.e. it has two perianth circles, two or one circle of stamens and one circle of carpels, and there are usually 5.4 or 2 members in a circle).

Therefore, for *Magnoliaceae*, the formulas $Ca_5 Co_5 A_{5+5} G_5$ are frequent. Lily-shaped flowers are often five-circular, three-membered; typical formula $Ca_3 Co_3 A_{3+3} G_3$. But there are many exceptions to this rule. So, in *Magnoliopsida* - *Berberidaceae*, *Lauraceae* and others - circles are three-membered, and in some *Liliopsida* and *Potamogeton* - circles of four-membered.

6. Representatives of these two classes form microspores and they differ in the order of arrangement. *Magnolioids* have a simultaneous type of microspore formation - four microspores are located in a tetrahedron, in *Liliaceae succinate* - all four microspores lie in one plane. Exception to this rule: In *Dicotyledons*: *Magnoliaceae*, *Ceratophyllaceae*, and in *Monocots*: *Orchidaceae*, *Iridaceae*, *Juncaceae*, etc.

Comparison of the features of the two classes showed that they cannot be separated by only one feature, but should be used and based on a complex of features. Some scientists - taxonomists, authors of textbooks (Kuznetsov, Bush, Lotsi) do not recognize *Monocotyledons* (*Liliopsida*) as a separate class and place them in different parts of the *Dicotyledonous* (*Magnoliopsida*). However, this division does not reflect the truth, since *Monocots* are a kind of group that differs in many ways from *Dicots*. For the first time the systematic significance of

monocotyledonous and dicotyledonous was recognized by D. Rey (XVII century), and then F. Brown, R. Varomling, A. Engler, R. Wettstein, I. Gorozhankin, M. Golenkin and A. Takhtajan in their phylogenetic systems divided the section Magnoliophyta into two classes: Monocotyledonous (Liliopsida) and Dicotyledonous (Magnoliopsida).

For many years, in the textbooks taught in higher education by P.M. Zhukovsky, M.V. Kultiasova, M. Golenkina, B.M. Kozopoliansky, L.I. Kursanov and others, in the textbooks of botany by S.A. Shostakovsky, S.S. Sakhobidinova the division of flowering plants are divided into the classes Monocotyledonous and Dicotyledonous.

Currently, many taxonomists consider the class Dicotyledons to be primary, and Monocots to be descended from the order Polycarpicales. (R. Wettstein, A. Takhtajan). A. Engler considered both classes to be parallel groups, although he placed Monocotyledons in front of Dicotyledons. Classic taxonomists A. Engler and R. Wettstein consider Monocots to be a monolithic group of monophilic origin. And according to G. Gallier, and N. Kuznetsov, monocots belong to a polyphyletic group. According to the latest information, at the present time on the globe Magnoliaformes are represented by two classes with 533 families, 13 000 genera and approximately 250 000 species. Based on modern scientific data (A. Takhtadzhyan, 1987), the evolution of higher plants can be represented by the following scheme:

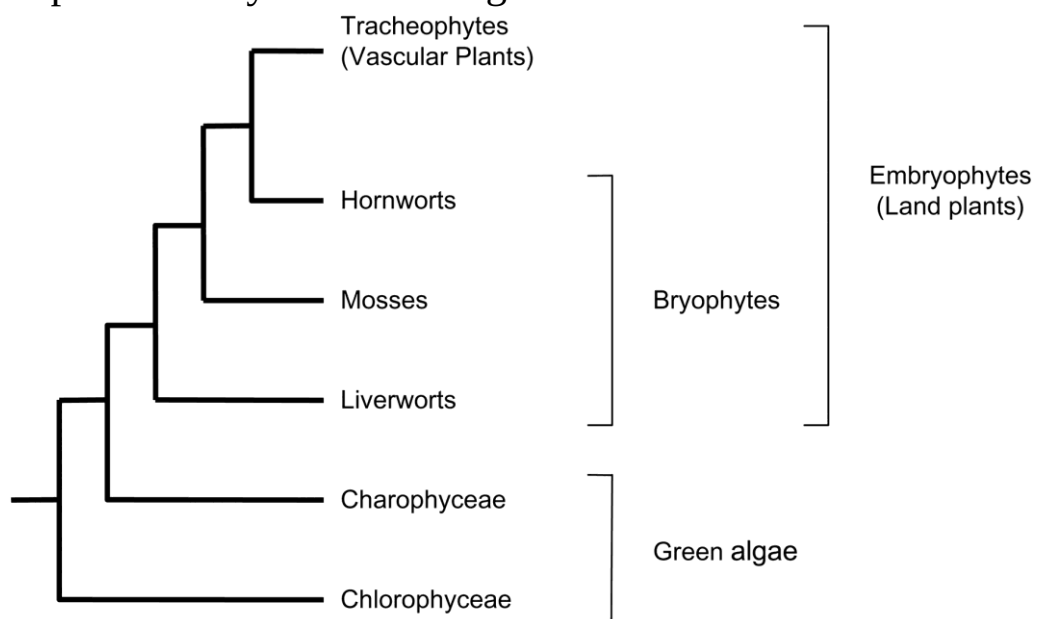


Figure 16. Evolutionary development of higher plants

9-CLASS

TOPIC: STRUCTURE AND REPRODUCTION OF GAMETOPHYTE AND SPOROPHYTE OF MAGNOLIOPHYTA (ANGIOSPERMATOPHYTA)

Demonstration materials and equipments: bud of a lily or amaryllis (or other plant), their flower, micropreparations of pollen growth and the structure of the ovule; drawing-schemes of double fertilization, micro and macrosporogenesis; microscopes, petri dishes, dissecting needles, and razor.

Take the following actions:

1. Study the structure of the anther and male gametophyte of a flowering plant on a microsample.

To do this, you need to take only opening flowers (lily or amaryllis) growing in the greenhouse and make a cross section of them or use a ready-made preparation. Find out how many cells the boot consists of? Each cell contains one sporangium. Find the wall of the cell on the preparation and the archesporia in the center. You will see that the wall of the sporangium consists of 4 layers: the epidermis, the fibrosis of the intermediate and the tapetum, which is in contact with the archesporium. Sketch the cut structure into the album.

Then it is necessary to examine a cross-section of the anther made from a fully opened lily or amaryllis flower. You will see that the inner layer of the sporangium tapetum and the intermediate layer dissolve and turn into a mucous substance. This substance is food for the growth of pollen. The cells contain a lot of pollen. They are formed as a result of the successive division of the archesporial cell, and from each as a result of reduction division, haploid tetrad cells are formed. In the anther, pollen grows and forms a gametophyte, within which a generative cell is formed, which subsequently gives rise to two sperms. The pollen is covered with intine and exine. Sketch the external structure of the pollen. On the finished preparation, study and sketch the ingress of pollen on the stigma and fertilization.

2. Study the structure of the ovule on the finished preparation.

In a microscope, you can see the structure of the ovule of a lily or tulip. First, sketch the structure of the ovule from a book or table. After that, you can easily distinguish between their structure and parts in a microscope. Find the pollen tube and ovule on the specimen, the type of ovule location (correct and incorrect). Find the embryo sac and the surrounding nucellus, as well as ovum, synergies, central cell and antipodes. Sketch the ovule and mark all its parts (Fig. 17).

3. Read, learn, make a summary, and sketch the diagram regarding "Microsporogenesis and the development of the male gametophyte", "The structure of the ovule and the development of the female gametophyte", and the theme: "The structure of the flower" from the book "Morphology of plants".

Based on the information gathered, sketch a diagram of the development cycle of Magnoliophyta.

4. In the order of the description of flowering plants, when identifying flowering plants, pay attention to the following:

A) LIFE FORM

-- tree, shrub, dwarf shrub, grasses (perennial, biennial, annual);
class - magnolia-shaped (dicotyledonous), and lily-shaped (monocotyledonous)

B) ROOT SYSTEM

- main root, adventitious roots, lateral roots.
- form: (biotype), main root, adventitious root, root sucker, branched.

- metamorphoses: root tubers, root crops, tubers.

C) STEM

- upright, creeping, curly,
- branched, not branched.
- pubescent, not pubescent
- cut: round, tetrahedral, triangular, flat.
- metamorphoses: rhizome, bulb, tubers, thorn

D) LEAVES

- simple: petiolar, sessile, vaginal; with stipules, without stipules, main swollen bell, shape of a leaf blade; lamina margins: whole, fingerlike, dentate; venation: parallel, arcuate, reticulate.

- compound without stipules, with stipules; leaf shape: leaf shape; leaf edges - whole, serrate, dentate, obtuse, number of leaflets,

- leaf arrangement: alternate, opposite, whorled, around the root collar (basal).

- metamorphoses: thorns, antennae, phyllodia

E) INFLORESCENCE

- monopodial: head, earring, ear, brush, umbrella, simple ear, basket; complex: - complex ear, complex umbrella.

- sympodial: monochasium (curl), dichasia, pleiochasium.

- no inflorescence (flowers are located singly)

F) FLOWER.

- actinomorphic, zygomorphic,

- perianth: complex, simple (calyx, corolla without perianth,

- calyx: loose, accrete (degree of connection, shape),

- corolla: free, accrete (degree of connection form); number of petals, color, presence of tubercles, presence of nectaries.

- stamen: the length is the same or not the same, the number,

- pistil: simple (apocarpous, syncarpous, number of carpels), complex (number of carpels)

- in the pistil: upper, lower, semi-lower ovary, number of stigmas,

- make a flower formula (Ca, Co, A, G)

G) FRUIT.

- simple, multi-seeded dry; leaflet, bean, pod, box; single-seeded dry: nut, **achene**, lionfish, weevil, acorn; multi-seeded juicy: berry, apples, pumpkin, drupe orange.

- complex: complex multi-leaf, complex multi-seed, complex multi-lattice, complex multi-leaf.

- compound fruit

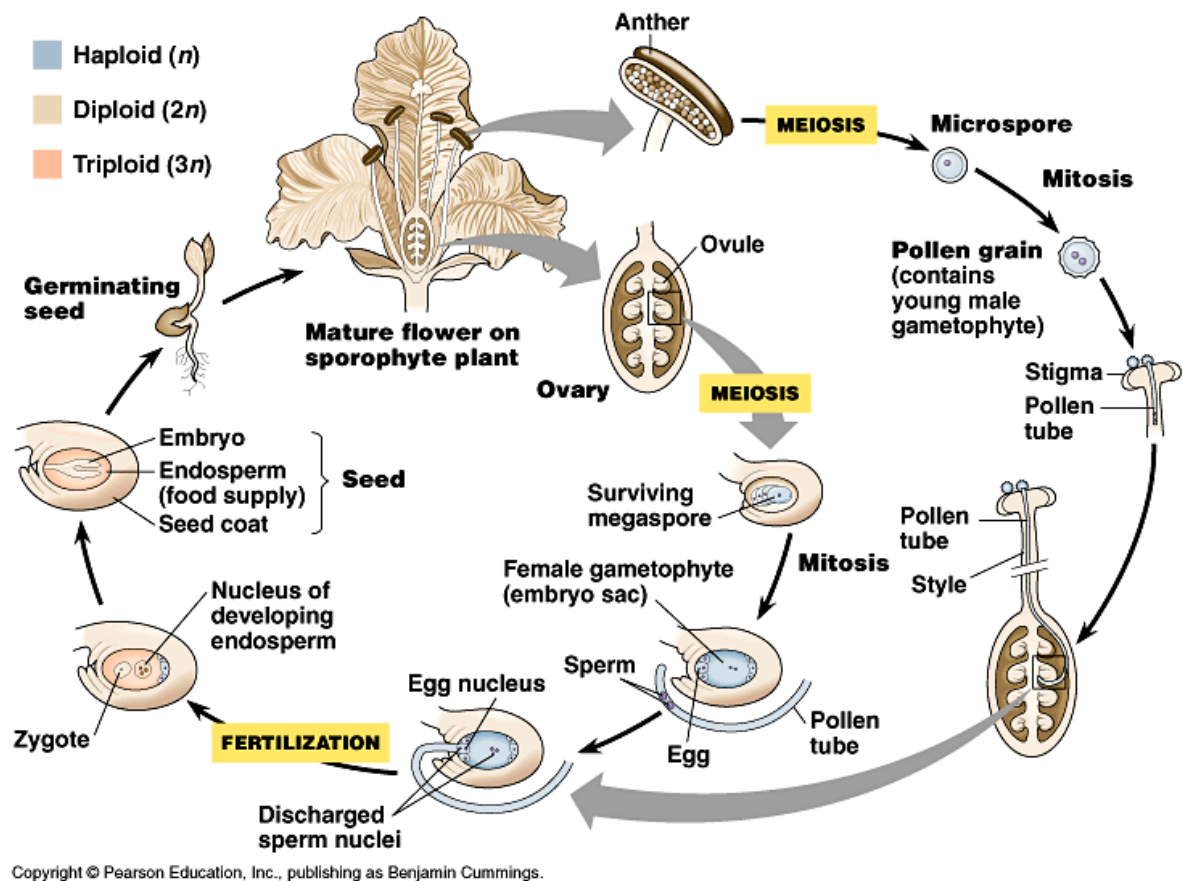


Figure 17. Development cycle of flowering plants

10-CLASS

TOPIC: STRUCTURE AND DEVELOPMENT CYCLE OF *MAGNOLIA GRANDIFLORA*

Division: *Magnoliophyta*

Class: *Magnoliopsida (Dycotyledones)*

Subclass: *Magnoliidae*

Order: *Magnoliales*

Family: *Magnoliaceae*

Subfamily: *Magnolideae*

Genus: *Magnolia*

Species: *Magnolia grandiflora*

Purpose of the class: to study the structure, development stages and classification of *Magnolia grandiflora*.

Objectives of the class: to teach students herbarium specimens, and to inform them about live materials and classification of the target plant; to strengthen the student's knowledge of the subject; to be able to answer the teacher's questions.

Demonstration materials and equipments: powerpoint slides; animations, online materials. Sketch of reproduction cycle of *M. grandiflora*, herbarium specimens, plant exhibits, prepared microscope slides (antheridium, archegonium, sporogonium), microscopes, magnifying glass, microscope slides, cover slips, preparation needles, water dropper (pipette).

LEARNING PROCEDURE

Students will get to know the herbarium specimens and exhibits of *Magnolia grandiflora* and learn about their biological characteristics using the teaching materials and books.

According to modern concepts, the Magnolia-like order is of the most ancient plants. This is evidenced by their anatomical and morphological structure, especially the shape of the receptacle and the order of arrangement, the shape of the parts of the flower.

To study the topic, do the following:

1. Get acquainted with the structure of *M. grandiflora* on herbarium or live material. You have familiarized yourself with the procedure for studying plants above. Give a characteristic of the plant, according to the drawing and the herbarium, and tell about the structure of the stem and its branches (pyramidal, oval, spreading) and write them down. Describe its shape, dimensions, surface and edges on the sheet. Study the structure of a flower using herbarium and living materials. While studying parts of perianth, pay attention to the shape of the receptacle (cone-shaped). The flowers are large, up to 10 cm, the petals are white, yellow, and pink in the amount of 6-12, and form 3-membered circles. The filaments of the stamens are extended, the pistils are almost free and are located spirally on the receptacle. Determine the

order of the stamens by the traces on the stem of the infructescence, and the gynoecium by the location of the fruits on the cone-shaped receptacle.

2. Having studied the perianth parts, the order of the stamens on the receptacle and the order of the pistil ovules, compare them with the scaly leaves of a pine. Find the phylogenetic affinity between flowering plants and gymnosperms.

Sketch a magnolia flower in the album, the location of the flower parts, and the structure of the fruit. Compose and write down the flower chart and formula. The formula of a large-flowered magnolia flower is $\text{Ca}_3 \text{Co}_{6-12} \text{A}_\infty \text{G}_\infty$ (Fig. 18).



Figure 18. Flower, fruit and seeds of *Magnolia grandiflora*

The second family of *Magnoliopsida* is the *Liriodendroideae* and it includes *Liriodendrom tulipifera* which is an ornamental plant. Study its structure on the herbarium material. Pay attention to the peculiar structure of the leaf - it is four-bladed. The number of stamens and pistils is numerous. Draw a shoot with leaves and flowers. Using books and lecture notes, write in a notebook the history of the appearance of magnolia and tulip trees in Uzbekistan, as well as modern information

about their places of growth and economic significance. Write down the names of the magnolia species growing in Uzbekistan in a notebook. Thus, despite the fact that *Magnoliaceae* is the first representative of flowering plants, the plan structure of the flower and wood is similar to gymnosperms.

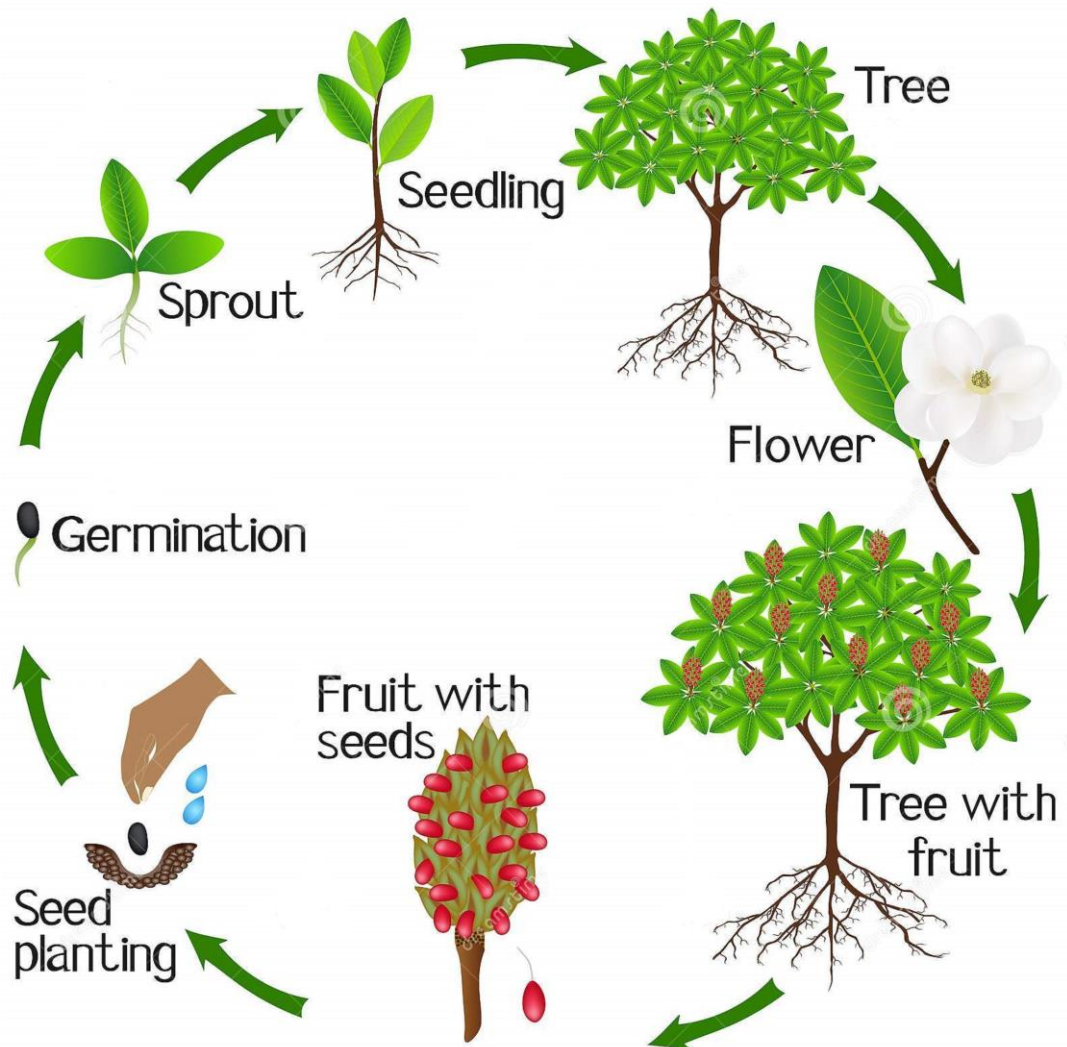


Figure 19. Life cycle of *Magnolia grandiflora*

Self Testing Questions:

1. Study the flower structure of *Magnolia grandiflora*.
2. Explain the classification of the plant and its origin.
3. Describe the morphological structure of *M. grandiflora*.
4. What is the reproduction cycle of *M. grandiflora*?
5. Draw the flower diagram of *M. grandiflora*.

11-CLASS

TOPIC: STRUCTURE AND REPRODUCTION OF *RANUNCULUS ACRIS*

Division: *Magnoliophyta*

Class: *Magnoliopsida*

Order: *Ranunculales*

Family: *Ranunculaceae*

Genus: *Ranunculus*

Species: *Ranunculus acris*

Purpose of the class: to study the structure, development stages and classification of *Ranunculus acris*.

Objectives of the class: to teach students herbarium specimens, and to inform them about live materials and classification of the target plant; to strengthen the student's knowledge of the subject; to be able to answer the teacher's questions.

Demonstration materials and equipments: powerpoint slides; animations, online materials. Sketch of reproduction cycle of *R. acris*, herbarium specimens, plant exhibits, prepared microscope slides (antheridium, archegonium, sporogonium), microscopes, magnifying glass, microscope slides, cover slips, preparation needles, water dropper (pipette).

LEARNING PROCEDURE

Students will get to know the herbarium specimens and exhibits of *Ranunculus acris* and learn about their biological characteristics using the teaching materials and books.

There are currently 66 genera, more than 2,000 species of the genus, and they are studied in five subfamilies. One of the central families is the *Ranunculus* family. Perennials are herbaceous plants, whose families differ sharply in the structure of the flower. Get acquainted with the available herbariums.



Figure 20. General view of *Ranunculus acris*

1. Investigate the diversity of the structure of the *Ranunculaceae* family growing in the flora of Uzbekistan:

- The flowers with simple perianth - *Anemone*, *Clematis*, *Thalictrum*.
- The flowers with compound perianth - *Ranunculus*, *Aquilegia*, *Ceratocephalus*.
- zygomorphic flowering groups - *Delphinium*, *Aconitum*.

2. Describe the structure of a flowering plant (*R. acris*) or a reptile (*R. repens*) according to the order in which flowering plants are described. Pay special attention to the vegetative organs and flower structure.

3. Draw an overview of the plant you are studying, a picture of the fruit and leaf, and a formula and diagram of the flower.

4. Identify 3-4 species of plants in the family and write the sequence of their identification in your notebook.

5. Get acquainted with the cultivated plants of the genus *Asteraceae* - *Paeonia*, *Nigella* and their importance.

Ranunculus acris is a perennial short-stemmed grass with claw-shaped leaves, 30-80 cm long, growing in meadows, swamps and streams. Stems are erect, cylindrical, branched, and hollow, the leaves alternate and are simple, claw-shaped, rhombic-lanceolate. The leaves around the root collar are long-stalked, the stems are short-stalked, sessile, without leaves.

Self Testing Questions:

1. How many genera and species are present in *Ranunculaceae*?
2. What are the representative species of *Ranunculaceae*?
3. In what habitats does *Ranunculus acris* grow?
4. Describe the morphological structure and the system of reproduction of *R. acris*.

12-CLASS

Topic: STRUCTURE AND DEVELOPMENT OF *ROSA CANINA*

Division: *Magnoliophyta*

Class: *Magnoliopsida (Dycotyledones)*

Subclass: *Rosidae*

Order: *Rosales*

Family: *Rosaceae*
Genus: *Rosa*
Species: *Rosa canina*

Purpose of the class: to study the structure, development stages and classification of *Rosa canina*.

Objectives of the class: to teach students herbarium specimens, and to inform them about live materials and classification of the target plant; to strengthen the student's knowledge of the subject; to be able to answer the teacher's questions.

Demonstration materials and equipments: powerpoint slides; animations, online materials; herbarium specimens and green individuals of the species from the genus of *Rosa*, *Medicago*, *Vexibia*, *Astragalus*, *Onobrychus*, *Trifolium*, *Alhagi*, *Glycyrrhiza*, *Psoralea*, *Trigonella*, *Melilotus*; fresh individual plants obtained in the spring season.

LEARNING PROCEDURE

Students will get to know the herbarium specimens and exhibits of *R. canina* and learn about their biological characteristics using the teaching materials and books.

Rosaceae. This family includes more than 3000 species belonging to about 100 genera. They are widespread on Earth. Depending on their life form, they include grasses, shrubs, bushes and trees. The leaves are arranged in a series, simple or complex, and oblong. Inflorescences are different. The flowers are straight, bisexual, mostly intricate inflorescence - inflorescences 5 (sometimes 4-6). The petals are not connected, the number of pollinators vary, usually many. The flower grows flat, plate-like or bulging, sometimes attached to the fruit, forming a false fruit. The seed consists of one or more petals -- knot top, half bottom or bottom.

The flower formula is $Ca_5 Co_5 A_\infty G_{(1-5)-\infty}$. The fruits are varied, including kernels, nuts, pistachios, canned fruits, fake fruits, berries and more. Representatives of the family are mainly propagated by seeds,

there are also those that propagate vegetative (e.g., roses, strawberries, etc.).

In Central Asia, 266 species of the family of 32 genera grow, and in Uzbekistan, 96 species of 11 genera.

The family is divided into 7 subfamilies according to the characteristics of generative and vegetative members. Among them which are widely represented in Uzbekistan are Spiraeoideae, Rosaideae, Maloideae-Pomoideae, and Prunoideae. The genus *Spiraea* includes mainly shrubs. The leaves are simple or complex, the flowers are shield-shaped or in clusters, the petals are complex, the petals are 5 in number. The fruit is a ball or pod. In Uzbekistan, 5 species grow mainly in the mountains. Some species are valued as medicinal, ornamental and honey-rich plants.

Sorbaria olgae is included in the Red List of the Republic of Uzbekistan.

Rosaideae subfamily includes the genera of *Rubus*, *Rosa*, *Hulthemia*, *Fragaria*. The category consists of thorny shrubs 1-3 m tall. The leaves are odd, lanceolate, complex, oblong. The flowers are **actinomorphic**, bisexual, large, white-pink, yellow, inflorescence complex. Inflorescence and petals 5 pcs. There are many pollinators and seeders. The fruit is a fake fruit (Fig. 21).

The genera includes about 120 species. Of these, 13 species grow naturally in Uzbekistan and several species are cultivated as ornamental plants. Among the roses, *Rosa canina* is widespread in Uzbekistan. The fruit of *R. canina* is used as a medicinal raw material, as a graft for grafting roses from seedlings grown from seeds. The *Rosaceae* family includes *Pyrus*, *Malus*, *Cydonia*, *Cotoneaster*, *Crataegus*, which are widespread in Uzbekistan and are closely related to the national economy.



Figure 21. Morphological parts of *Rosa canina*.

A- reproductive rod; B-flower, V-longitudinal section of flower, G-complex ball (hypothalamus); D- flower stalk, E - flower diagram, J-pollinator, Z- seeder, I- reproductive rod.

Self Testing Questions:

1. How many genera and species are present in *Rosaceae* family?
1. Count the representative species of *Rosaceae* family?
2. In what habitats does *Rosa canina* grow?
3. What is the morphological structure of *Rosa canina*?

13-CLASS

TOPIC: STRUCTURE AND DEVELOPMENT OF *MEDICAGO SATIVA*

Class: *Magnoliopsida (Dicotyledoneae)*

Subclass: *Rosidae*

Order: *Fabales*

Family: *Fabaceae*

Subfamily: *Faboideae*

Species: *Medicago sativa*

Purpose of the class: to study the structure, development stages and classification of *Medicago sativa*.

Objectives of the class: to teach students herbarium specimens, and to inform them about live materials and classification of the target plant; to strengthen the student's knowledge of the subject; to be able to answer the teacher's questions.

Demonstration materials and equipments: powerpoint slides; animations, online materials. Herbarium specimens and green individuals of the species from the genus of *Pisum*, *Medicago*, *Vexibia*, *Asrtagalus*, *Onobrychis*, *Trifolium*, *Alhagi*, *Glycyrrhiza*, *Psoralea*, *Trigonella*, *Melilotus*; fresh individual plants in the spring season.

LEARNING PROCEDURE

The class, on studying the *Medicago sativa*, begins by introducing the herbariums, exponents and alive plant materials.

Fabaceae, which is the third largest family among the angiosperms after *Orchidaceae* (orchid family) and *Asteraceae* (aster family), consists of more than 650 genera and about 18,000 species and is worldwide in distribution. The leaves usually are pinnately compound (feather formed), sometimes trifoliate (with three leaflets), or palmate (the leaflets radiating from a common point). The leaves of a few species are simple or reduced to scales. The fruit is typically a legume,

or pod, which splits open as it dries, releasing the seeds. Symbiosis of nodulating bacteria at the root is an important characteristic feature of the family. Identification of genera within the family, the location of stamen is very important. In the *Vexibia*, *Thermopsis*, all 10 stamen are free; in *Lupinus* and *Genista*, 10 of the stamen grew together with the filaments and formed a pollen tube. However, in most groups, 9 of the pollen grains are mixed with yarn, and 1 of them is *Pisum*, *Cicer*, *Medicago*, *Trifolium*, *Alhagi*.

In *Vexibia* and *Thermopsis* genera, all 10 stamen are free; In *Lupinus* and *Genista*, 10 of the stamen also grew together with the filaments and formed a pollen tube. However, in most groups, 9 of the stamen are mixed with filament, and 1 of them is free - *Pisum*, *Cicer*, *Medicago*, *Trifolium*, *Alhagi*.

The fruit structure of the peppers is also unique. In some categories the pods are opened in two layers (mung, beans, peas, nuts), in others the pods are divided into multi-seeded but single-seeded pieces (*Coronilla*, *Hedysarum*), white in esparsets, the legume is single-seeded and it does not open.

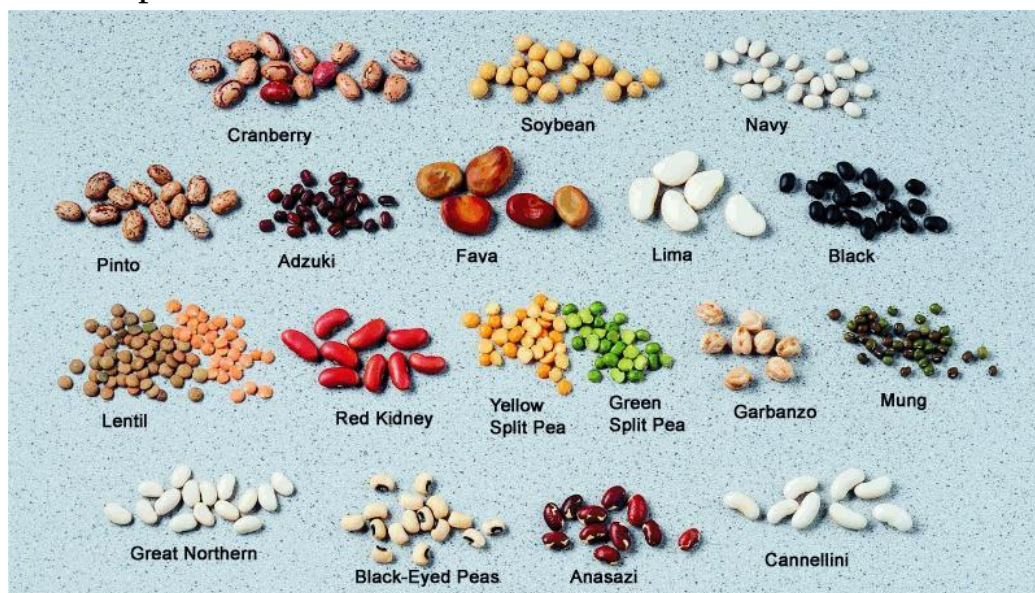


Figure 22. Different seeds of legumes

Write in your notebook the scientific and local names of several cultural species from the corners and point out their importance.

Medicago sativa is a perennial polycarp grass about 50-90 (100) cm tall. The root is a bullet root and has many wheat and millet grains.

Nitrogen-fixing bacteria *Azotobacter radiculicola* are located in the tubers and are responsible for fixing and accumulating atmospheric nitrogen. Stems erect, branched, rounded, ribbed. The leaves are trifoliate complex, linear; The leaves are elongated, serrated.

The flowers are densely clustered, longer than the leaves, zygomorphic, with a typical structure. Flower Castle 2nd floor: flower cup consists of 5 adjacent flower petals. The inflorescence consists of 5 flower petals, 3 of which are loose (sail, 2 oars), and 2 of which form a grip and a boat at the end. Of the pollinators, 9 are 4 + 5 in shape and form 2 rings, and 1 is free, located separately. Seeds 1, bent at the base, nodular top, single-celled, several-seeded. Flower diagram

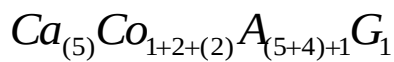


Figure 23. General view and morphological parts of *Medicago sativa*:

(a) flower; (b) fruiting; (c) fruit



Figure 24. *Medicago sativa*: flowers (top left), fruits (right top and bottom), seeds (left bottom)

Fruit-spiral twisted pods. After examining the herbarium, the leaf, the inflorescence, the flower; a picture of a pollinator, a seed, and a fruit is drawn. Then 2-3 wild species of alfalfa (*Medicago*) are identified. Keep notes of what you have learned from the process.

Self Testing Questions:

1. General description of *Fabaceae* family.
2. General morphological structure of *Medicago sativa*.
3. How many genera and species are included in *Fabaceae* family?
4. Describe the flower and fruit of *Fabaceae* family.

14-CLASS

TOPIC: STRUCTURE AND DEVELOPMENT OF *MENTHA ASIATICA*

Class: *Magnoliopsida (Dicotyledoneae)*

Subclass: *Lamiidae*

Order: Yalpiznamolar - *Lamiales*

Family: *Lamiaceae*

Subfamily: *Lamioideae*

Species: *Mentha asiatica*

Purpose of the class: to study the structure, development stages and classification of *Mentha asiatica*.

Objectives of the class: to teach students herbarium specimens, and to inform them about live materials and classification of the target plant; to strengthen the student's knowledge of the subject; to be able to answer the teacher's questions.

Demonstration materials and equipments: powerpoint slides; animations, online materials. Herbarium specimens and green individuals of the species from the genus of *Mentha*, *Salvia*, *Phlomis*; *Ziziphora*, *Origanum*, *Eremostachys*; fresh individual plants in the spring season.

LEARNING PROCEDURE

The class on studying the *Mentha asiatica* begins by introducing the herbariums, exponents and live plant materials.

The *Lamiaceae* is a family of flowering plants with a cosmopolitan distribution containing about 236 genera and has been stated to contain more than 7000 species. The largest genera are *Salvia* (900), *Scutellaria* (360), *Stachys* (300), *Plectranthus* (300), *Hyptis* (280), *Teucrium* (250), *Vitex* (250), *Thymus* (220), and *Nepeta* (200). Within the *Lamiaceae* family 11 subfamilies are differentiated. One of the central subfamilies of *Lamiaceae* is *Lamioideae* which includes 34 genera.

Most members of the family are perennial or annual herbs with square stems, though some species are woody shrubs or subshrubs. The leaves are typically simple and oppositely arranged; most are fragrant and contain volatile oils.

Flowers have a calyx of 5 united petals and a corolla of 5 united sepals. They are usually bilaterally symmetrical with a superior ovary. There are 4 stamens attached to the petals: 2 of them longer and 2 are shorter. However, *Salvia* and *Rosmarin* have only 2 stamen. Configuration of the anthers is an important characteristic in separating the genera. The ovary is composed of two carpels, each two lobed, so it almost appears as 4 united carpels. The style is two forked. The flower formula is: $Ca_{(5)}Co_{(2+3)}A_{2+2}G_{(2)}$.

The fruits are 4 small seedlike nutlets.

***Mentha asiatica*.** The most common perennial polycarp in Uzbekistan. Stems 60-70 (100) cm long, branched, erect, 4-sided, covered with fine silvery hairs, leaves elongated, the base is round or slightly heart-shaped; sessile or short-banded, with soft fine hairs, the leaf edge is serrated.



Figure 25. General view and the structure of *Mentha asiatica*

1-rootstock, 2-stem with inflorescence, 3-leaves, 4-location of stamen in petals, 5- the structure of pistil, 6-flowers, 7-fruit structure.

Inflorescences are formed at the tip of the stem or on the side branches that protrude from the leaf axils at the top of the stem. Equal to or slightly shorter than the corolla, they form a ring in the common corolla. The petals are hairy and fluffy. The cup is light pink 2.0-2.5 (3) mm, thick hairy, the teeth are 1.5-2.0 (3) times shorter than the crown tube. The crown is 3-4 mm, whitish pink.

It flowers in June-August and bears fruit in July-September. It grows along the streams, canals, mountain streams, and wetlands.

Three species of mint are found in the native flora of Uzbekistan.

At the end of the lesson, identify a few species of mint. Find out and write the importance of cultivated varieties.

Self Testing Questions:

1. General description of *Lamiaceae* family.
2. Morphological structure and features of *Mentha asiatica*.
3. How many genera and species are included in the *Lamiaceae* family?

15-CLASS

TOPIC: STRUCTURE AND DEVELOPMENT OF *TARAXACUM OFFICINALES*

Class: *Magnoliopsida (Dicotyledoneae)*

Subclass: *Asteridae*

Order: *Asterales*

Family: *Asteraceae*

Subfamily: *Asteroideae*

Species: *Taraxacum officinales*

Purpose of the class: to study the structure, development stages and classification of *Taraxacum officinales*.

Objectives of the class: to teach students herbarium specimens, and to inform them about live materials and classification of the target plant; to strengthen the student's knowledge of the subject; to be able to answer the teacher's questions.

Demonstration materials and equipments: powerpoint slides; animations, online materials. Herbarium specimens and green individuals of the species from the genus of *Taraxacum*, *Centaurea*, *Acroptilon*, *Inula*, *Achillea*, *Xanthium*, *Helianthus*, *Carthamus*; fresh single plants from the spring season.

LEARNING PROCEDURE

The class on studying the *Taraxacum officinales* begins by introducing the herbariums, exponents and alive plant materials.

Asteraceae family is a large family of Magnoliopsida class and occupies the highest position in the evolution of flowering plants. It is now estimated that there are 1250-1300 genera and about 20,000-25,000 species in the family, which is 1/10 of all flowering plants.

The distinctive feature of the family is that the inflorescence is in the form of **capitulum** or florets. The smaller florets, in turn, form a complex shield or broom-like inflorescence so called capitulescences (*Achillea*, *Handelia*, some species of *Artemisia*). Each capitulum is wrapped in several rows of wrapping leaves. The number, shape, size, and color of the rows of leaves are important systematic features in determining the series. The base of the capitulum can be flat, concave, convex, elongated, and it determines the shape of the capitulum.

The leaves of *Asteraceae* are simple or occasionally compound, and their arrangement along the stem may be opposite, alternate, or, less commonly, whorled; not infrequently they are opposite toward the base of the stem and alternate above.

The most obvious and outstanding general feature of *Asteraceae* is that the flowers are grouped characteristically into compact inflorescences (heads) that superficially resemble individual flowers. Each such head is ordinarily subtended by an involucre of small modified leaves (bracts). Furthermore, in more than half the members of

the family, the flowers in the outermost row or rows of the head have a modified, mainly flat and elongate corolla that resembles an individual petal of most other flowers. Thus, the “petals” of a daisy or sunflower are actually the outermost flowers of the head. An inflorescence of this family can have more than 1,000 individual flowers (florets), and the heads can be grouped into more complex, secondary arrangements called capitulescences.

When present, the flower petals of *Asteraceae* are joined together by their margins, forming a tubular or strap-shaped corolla (a sympetalous corolla) that often has apical teeth representing the petal tips. The other floral parts are attached to the top of the ovary rather than beneath it. The calyx (sepals) of *Asteraceae* is so highly modified, in contrast to that of other families, that it is given a different name, the pappus. The pappus consists of one to many dry scales, awns (small pointed processes), or capillary (hairlike) bristles; in some the scales may be joined by their margins to form a crownlike ring at the summit of the ovary. The flower formula is $Ca_{(ped.)}Co_{(5)}A_{(5)}G_{(2)}$.

The members of *Asteraceae* produce a type of fruit called an **achene**, which is dry and single-seeded and does not open at maturity. The apparent seeds of the sunflower, for example, are actually achenes. The seeds of many *Asteraceae* species are distributed in a variety of ways, often aided by modifications of the floral pappus. When the pappus consists of numerous capillary bristles, as in *Taraxacum officinale* (dandelion), it facilitates distribution of the achenes by the wind by providing buoyancy.



Figure 26. General view and morphological parts of *Taraxacum officinales*

In the class we can get to know the structure of *T. officinales* – one of the typical representative of *Asteraceae* family. *T. officinales* is a perennial herbaceous plant with tap root system penetrating up to 15-25 cm in depth. It is an almost cosmopolitan plant, very common in oases. The stems are very short and do not rise to the surface. The leaves are simple, sessile, sometimes entire, and bulge around the root collar. At the end of each flower grows a capitulum with a diameter of 3-4 cm. The leaves are in 3 rows, the outer leaves are narrow, linear, the tip is

bent down, 7-12 mm, the flower bed is smooth. All the flowers in the capitulum are 5-toothed, lanceolate, bisexual, golden-yellow, with 5 pollen grains and pollen grains forming a ring. Gynoecium is syncarp, consisting of 2 fruiting leaves, and an inferior ovary. The flower formula is $Ca_{(ped.)}Co_{(5)}A_{(5)}G_{(2)}$. The fruits are achenes. It flowers almost during the whole year and gives fruits. It is a very common weed of grassland and is cultivated. *T.officinales* is a commonly used herb for remedy. It is especially effective and valuable as a diuretic because it contains high levels of potassium salts and therefore can replace the potassium that is lost from the body when diuretics are used. At the end of the class draw the general diagram-along with its morphological parts in your album.

Self Testing Questions:

1. Give general description of *Asteraceae* family.
2. The structure and reproduction of *Taraxacum officinales*
3. How many genera and species are included in the *Asteraceae* family?
4. What are the specific features of *Asteraceae* family?

16-CLASS

TOPIC: STRUCTURE AND DEVELOPMENT OF *TULIPA GREIGII*

Class: *Liliopsida* (*Monocotyledones*)

Subclass: *Liliidae*

Order: *Liliales*

Family: *Liliaceae*

Species: *Tulipa greigii*

Purpose of the class: to study the structure, development stages and classification of *Tulipa greigii*.

Objectives of the class: to teach students herbarium specimens, and to inform them about live materials and classification of the target plant; to strengthen the student's knowledge of the subject; to be able to answer the teacher's questions.

Demonstration materials and equipments: powerpoint slides; animations, online materials. Herbarium specimens and green single plants of the species from the genus of *Tulipa*, *Gagea*, *Fritillaria*, *Rhinopetalum*, fresh single plants from the spring season.

LEARNING PROCEDURE

The class on studying the *Tulipa greigii* begins by introducing the herbariums, exponents and alive plant materials.

In modern botanical textbooks, the family Liliaceae is much larger, consisting of about 11 subfamilies. The Liliaceae is one of these families, named after Takhtajyan A.L. (1966, 1987) and is scientifically substantiated as a separate family. Liliaceae, the lily family of the flowering plant order Liliales, with 16 genera and 635 species of herbs and shrubs, is native primarily to temperate and subtropical regions. Members of the family are herbaceous perennials. The leaves usually have parallel veins and are clustered at the base of the plant but may alternate along the stem or be arranged in whorls. The leaves are simple, without bands, alternating at the base, and their edges are toothless. Most species have an underground storage structure, such as a bulb. The bulb is covered with a juicy fleshy skin.

The flowers are large, solitary (tulips) or sparsely inflorescence (gooseberry), red, yellow, white, and purple. The flower parts are arranged in four circles of three. The flower formula is $Ca_0Co_{3+3}A_{3+3}G_{(3)}$. Occasionally the fruits are berries.

In Uzbekistan, there are 4 genera of *Liliaceae* family.

***Tulipa greigii* Rgl.** The plant embodies all the characteristics of the structures of tulips. It is ephemeroid with ovoid or round bulbs 15-45 cm tall. The leaves are 3-4, sparse at the base, curved at the edges, with

dark red spots on the surface, and the bottom is broadly pubescent, elongated, without bands, 10-32 cm long and 3-10 cm wide. The flower is red, 4-5 cm long, orange-red, sometimes yellow, yellowish, the bottom is yellow and has ovoid black spots. The stamen are 6, the strip is black or yellow, the anther is yellow, purple, and gray (Fig. 27).

The capsule is straw-yellow, light brown, 11 cm long and wide up to 4 cm. Seeds are ovoid, light brown and 1-3 cm. It flowers in April-May and bears fruit in June-July. It grows in the foothills of the mountains in the clayish and gravelly soils. It is a rare species whose range of distribution is sharply declining. The plant is included in the Red Book of Uzbekistan.

Draw a diagram of the structure of the *T. greigii* and mention its parts.

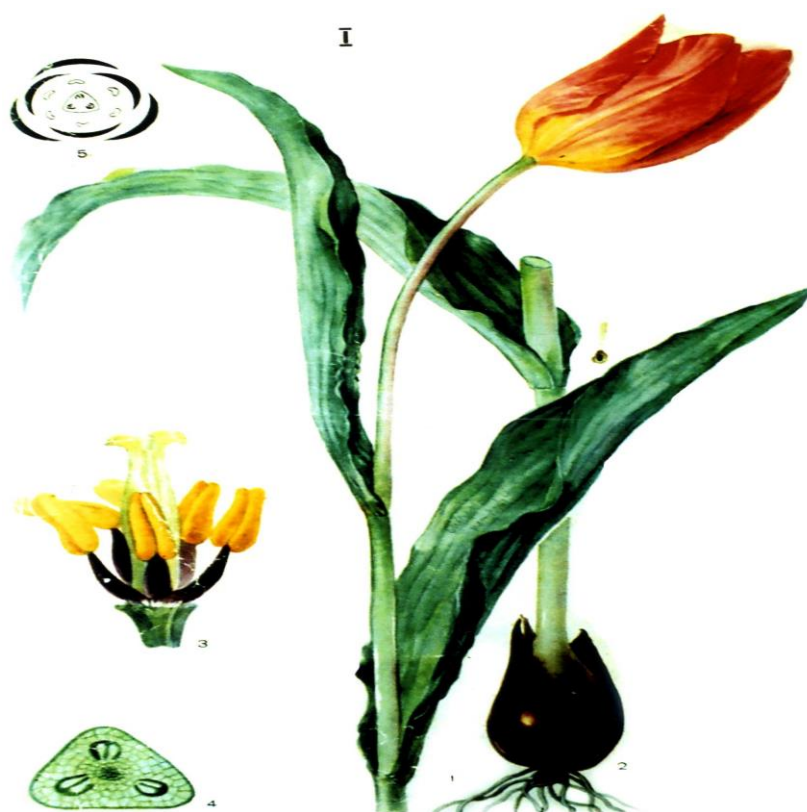


Figure 27. General view of *Tulipa greigii* Rgl

1-2 – general view, 3-flower without perianth, 4-cross section of ovary, 5-flower diagram.

Self Testing Questions:

1. General description of *Liliaceae* family.
2. The structure and reproduction of *Tulipa greigii*.
3. How many genera and species are included in the *Liliaceae* family?
4. What are the specific features of the *Liliaceae* family??
5. What is the flower structure of the *Liliaceae* family?
6. Describe the main characteristics of *Tulipa greigii*.

17-CLASS

TOPIC: STRUCTURE AND DEVELOPMENT OF *ALLIUM* *SUVOROVII*

Class: *Liliopsida* (*Monocotyledones*)

Subclass: *Liliidae*

Order: *Liliales*

Family: *Alliaceae*

Species: *Allium cepa*, *A. sativum*, *A. Suvorovii*

Purpose of the class: to study the structure, development stages and classification of *Allium cepa*, *A. sativum*, and *A. Suvorovii*.

Objectives of the class: to teach students herbarium specimens, and to inform them about alive materials and classification of the target plant; to strengthen the student's knowledge of the subject; to be able to answer the teacher's questions.

Demonstration materials and equipments: powerpoint slides; animations, online materials. Herbarium specimens and green individuals of the species from the genus of *Allium*, fresh individual plants from the spring season.

LEARNING PROCEDURE

The class on studying the *Allium cepa*, *A. sativum*, and *A. Suvorovii* begins by introducing the herbariums, exponents and alive plant materials.

Using the recently published teaching manuals, students should make notes on the main characteristics of the **Alliaceae** family and learn the obtained information. This family was separated from the *Liliaceae* family by A. Takhtadjyan and is distinguished from close families (tulips, lilies) by a number of features. This family contains no cardioglycosides, pungent-smelling allyldisulfides, the presence of milky or colorless volatile tubules on the leaves and bulbs, as well as the presence of umbrella-shaped inflorescences and curtain-like wraps.

There are 32 genera and 750 species of families on Earth. In the flora of Uzbekistan, *Alliaceae* has a single genus and 68 species. All bulbous, bulbous-root-stemmed ephemeral perennials.

The flowers are simple inflorescence, **actinomorphic**, bisexual, three-membered, and the pistil is simple and syncarp, while the ovary is superior. The fruit is a capsule with 3 partitions.

To get the class acquainted with herbariums of onions, garlic bulbs, or other wild onion species, and to use fixed flowers or tubers that grow in pots in the greenhouse too. The morphological structure and differences of *Allium cepa* and *A. sativum* are determined.

Allium cepa, commonly known as onion, belongs to the *Liliaceae* family (**Fig.28**). *A. cepa* is biennial herbaceous plant with bulb. The bulbs are composed of shortened, compressed, underground stems surrounded by fleshy modified scales (leaves) that envelop a central bud at the tip of the stem. The leaves are yellowish to bluish green and grow alternately in a flattened, fan-shaped swathe. They are fleshy, hollow, and cylindrical, with one flattened side. Inflorescence takes on a spherical umbel arrangement. The fruit is a capsule with 3 partitions, the seed is 2 mm and of black color.

A. sativum is a perennial flowering plant growing from a bulb. The leaf blade is flat, linear, solid, and approximately wide, with an acute apex. The plant may produce pink to purple flowers from July to

September. The bulb contains outer layers of thin sheathing leaves surrounding an inner sheath that encloses the clove. Often the bulb contains 10 to 20 cloves that are asymmetric in shape, except for those closest to the center. It produces hermaphrodite flowers. The flower formula is $\text{Ca}_0\text{Co}_{3+3} \text{A}_{3+3}\text{G}_{(3)} \text{yoki } \text{P}_{3+3} \text{A}_{3+3} \text{G}_{(3)}$. It is pollinated by bees, butterflies, moths, and other insects.

***Allium pskemense*.** It is a Perennial herb with a 40-150 cm height. The leaves are 3-6, cylindrical, 2-3 cm wide, and the top is slender. The flowers are small and white. It blooms in June and the fruits ripen in August. The fruit is a capsule. It grows only in the mountains of Tashkent region, on rocky substrate, in cracks of the rocks.

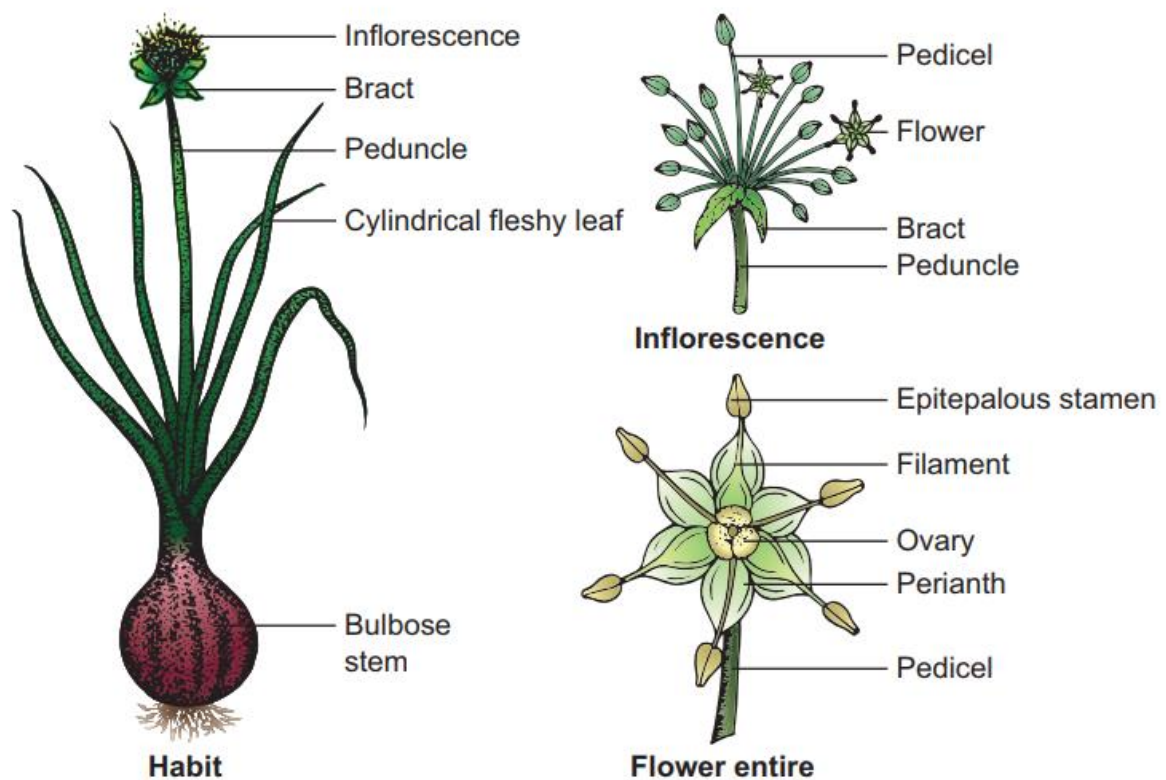


Figure 28. General view and morphological parts of *Allium cepa*

***Allium stipitatum*.** It is a perennial herb with 60-150 cm height. Its 4–6 basal leaves are broad, green to greyish green in colour, and is variably hairy. The leaves normally wither by the time the bulb

flowers. Flowers are borne on stems which are 60–150 cm (24–59 in) tall and arranged in an umbel. The petals are light purple. Onions are almost spherical, and their skin is dark. The fruit is a capsule. It blooms in May-June and ripens in July. The plant was included in the Red Book of Uzbekistan, but due to its regenerative nature, this plant was excluded from the Red Book.

Self Testing Questions:

1. General description of the *Alliaceae* family.
2. Structure and reproduction of *A. stipitatum*.
3. How many genera and species are included in the *Alliaceae* family?
4. What are specific features of the *Alliaceae* family?
5. What is the structure of flowers of this family?
6. Describe the general characteristics of *A. cepa*.
7. What is the structure of *A. sativum*?
8. What is the flower formula of *A. cepa*?

18-CLASS

TOPIC: STRUCTURE AND DEVELOPMENT OF *TRITICUM AESTIVUM*

Class: *Liliopsida (Monocotyledoneae)*

Subclass: *Liliidae*

Order: *Poales*

Family: *Poaceae*

Subfamily: *Poaideae*

Species: *Triticum aestivum*

Purpose of the class: to study the structure, development stages and classification of *Triticum aestivum*.

Objectives of the class: to teach students herbarium specimens, and to inform them about alive materials and classification of the target plant; to strengthen the student's knowledge of the subject; to be able to answer the teacher's questions.

Demonstration materials and equipments: powerpoint slides; animations, online materials. Herbarium specimens and green individuals of the species from the genus of *Triticum*, *Hordeum*, *Poa*, *Agropyrum*, *Taeniatherum*, *Aegilops*, *Bromus*, and *Zea*.

LEARNING PROCEDURE

The class on studying the *Triticum aestivum* begins by introducing the herbarium, exponents and alive plant materials.

Poaceae family is the largest species-rich family of the class Liliopsida. Currently, this family includes about 11,000 species belonging to 900 genera. In most cases, it forms the main part of the vegetation cover (steppe, desert). They are the most important food and fodder plants, and are widely distributed on the Earth. Most species are represented by grasses, and in the tropics there are some tree species (bamboos) too.

Grasses are annual, biennial, or perennial plants that are usually herbaceous but may be woody in some genera. They may be terrestrial or aquatic. The leaves may be evergreen or deciduous and are all basal or alternate, and usually much longer than wide. The leaves have parallel veins and consist of a cylindrical sheath that wraps around the stem with a blade above it. The stems have conspicuous nodes where the base of the leaf attaches them. The flowers are generally small, without obvious sepals or petals, and are arranged in structures called spikelets that consist of the flowers with specialized subtending scales (called glumes, lemmas, and paleas). There are from 1-3 stamens and 2-3 styles topped with branched stigmas designed to capture pollen from the air. The fruit is usually a caryopsis (i.e., a grain).

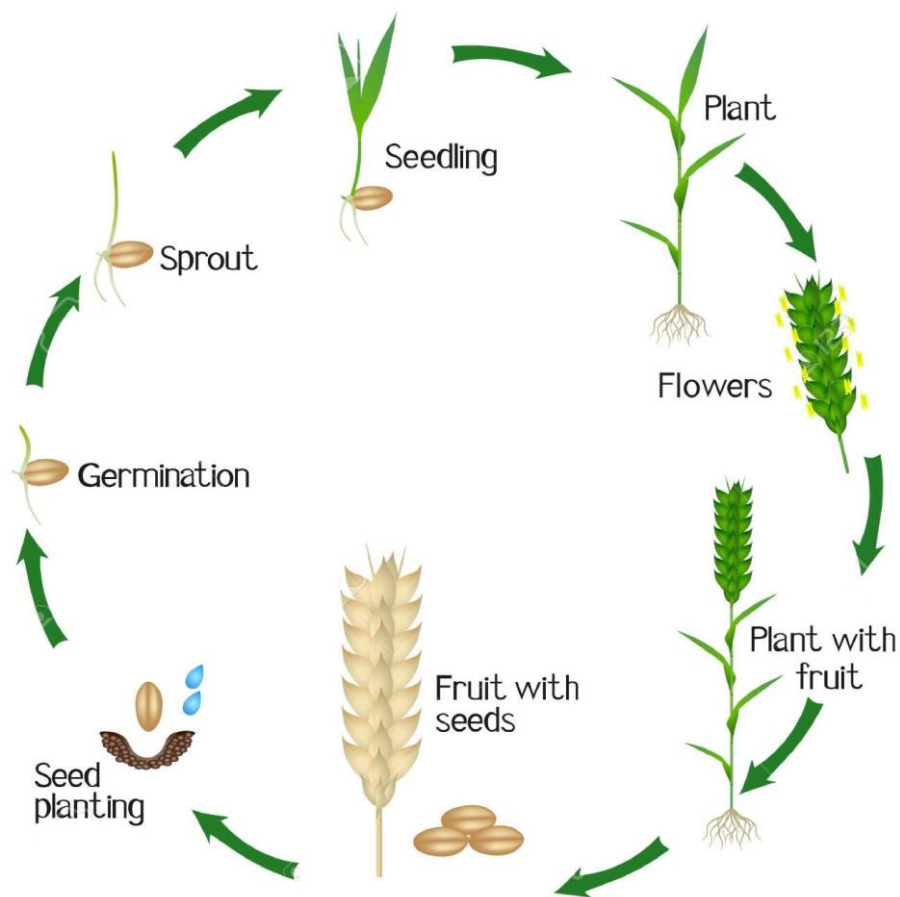


Figure 29. Development cycle of *Triticum aestivum*

The stems are hollow or pithy with straw stems, sometimes the center of the stem is filled with a soft tissue (in corn, sorghum), divided into nodes. The roots are fibrous, with an adventitious root system. The leaves are simple (except bamboo), flat and narrow, parallel veined, arranged alternately on the stem. The leaf collar is broad and auricles are small and hairy. Ligule is rounded and appears frayed. Leaves may have hairs.

The flowers are small, colorless, without inflorescences and clustered in short, simple inflorescences. Slender stalks produce flowers. Flowering begins at the middle third of the spike and continues towards the basal and apical parts in 3 - 5 days. All spike-bearing tillers eventually flower almost simultaneously. Wheat is normally wind-pollinated.

Flowers that are fertilized produce seeds called grains or kernels. The wheat kernel is made up of three parts: the endosperm, the bran, and

the germ. The tiny germ is the part of the kernel that will sprout and grow into a new wheat plant if the seed is planted.

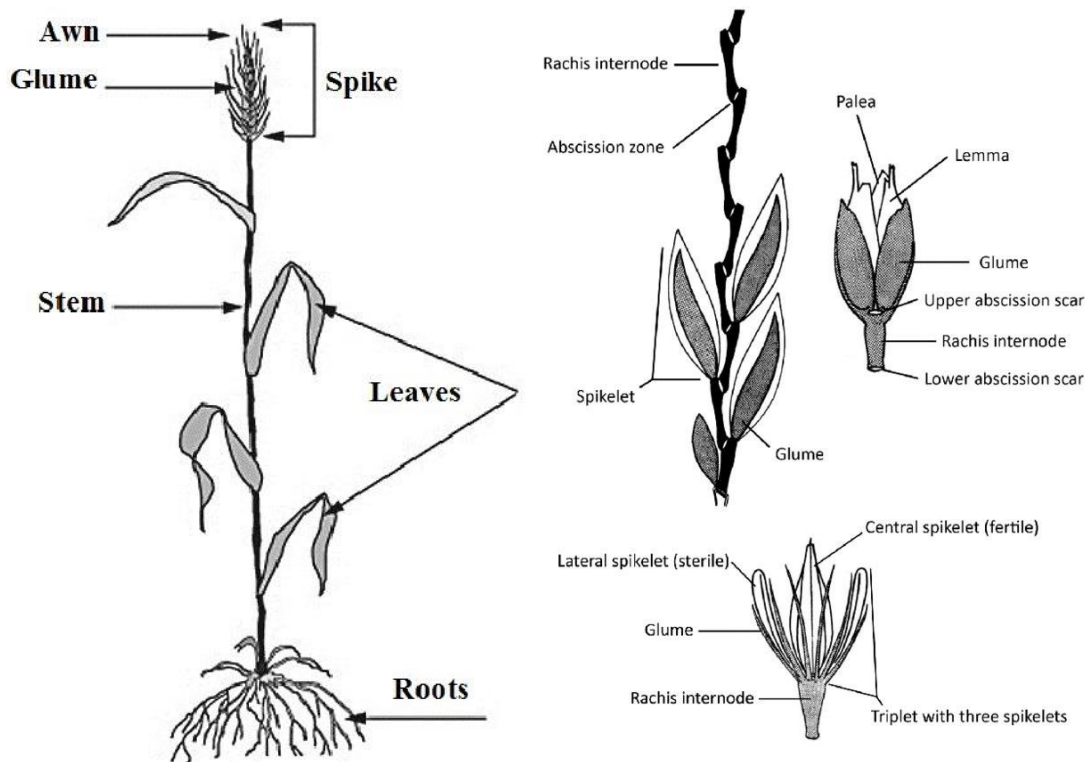


Figure 30. Morphological parts of *Triticum aestivum*

The structure of the *Poaceae* family will be analyzed in the example of wheat. Wheat is an annual, biennial (autumn) hollow stem plant with inflorescence and a compound spike. The **spikelets** have 3-7 flowers and protrude from the curves of the **rachis**. The rachis is curved, angular, unbreakable (in cultivars) or are fragmented (in wild species). The spikelet is 2-toothed. The lower (outer) flower stalk is long-edged or serrated, while the upper (inner) is curtain-shaped. The pollen grains are 3 lobes, the seed pods are 25 lobes and pinnate, the lobes are 2 lodicles. It is mainly pollinated spontaneously and by wind. The fruit is free-falling.

There are 19 species of wheat, 13 of which are found in the Transcaucasian region and 4 in Central Asia. More than half of the world's bread is made from wheat. All cultivated wheat varieties are hard wheat (*T. durum* L.) and soft wheat (*T. aestivum* L.), especially soft wheat is widely distributed.

Take a grain of wheat from the herbarium and study it carefully. Describe the length of the grain, the number of grains, the number of grains in the spike, the size of the grain, its color, and write it in your notebook.

Self Testing Questions:

1. Give the general characteristics of *Poaceae* and *Cyperaceae*.
2. Explain the structure and reproduction of *T. aestivum*.
3. How many genera and species do *Poaceae* and *Cyperaceae* have?
4. Describe the morphology of *T. aestivum*.

BOTANICAL TERMS
used in the methodological manual on "Botany. Systematics of Higher Plants"

English	Uzbek	Russian
Aboveground part	yer ustki qismi	надземная часть
absorption (root function)	absorbsiya	абсорбция
accessory fruit	soxta meva	плод ложный
achene	danakcha	семянка
actinomorphous, actinomorphic	aktinomorf	актиноморф
alder	olxa	ольха
alternate	navbat bilan	очередное
alternation of generations	nasllarning gallanishi	чередование поколений
amphigastrium (<i>pl. amphigastria</i>)	amfigastriya	амфигастрий
anchorage (root function)	mustahkamlovchi, biriktirib turuvchi	закрепление
androecium	androtsey	андроцей
angiosperms	yopiq urug‘li o‘simliklar	покрытосеменные растения
annual	bir yillik	однолетний
anther	changdon	пыльник
antheridial stalk	anteridiyli tirgovuch	антеридиальная подставка
antheridiophore	anteridiyli tallom	антеридиальный таллом
antheridium (<i>pl. antheridia</i>)	anteridiy	антеридий
anthocyanins	antotsian (pigment)	антоциан - синий цвет
aquatic environment	suvli muhit	водная среда
archegonic stalk	arxegoniyli tirgovuch	архегониальная подставка
archegoniophore	arxegoniyli tallom	архегониальный таллом

archegonium (<i>pl. archegonia</i>)	arxegoniy	архегоний
areole	shakli o'zgargan yon kurtak (kaktuslarda)	видоизменённая боковая почка (у кактуса)
asexual phase (<i>sporophyte</i>)	jinssiz bo'g'in (<i>sporofit</i>)	бесполоя фаза (<i>спорофит</i>)
awn (in wheat)	qiltiq (bug'doyda)	ость (в пшенице)
B eech	buk, qora qayin	бук
beet	lavlagi	свекла
belowground part	er ostki qismi	подземная часть
berry	rezavor meva	ягода
biennial	ikki yillik	двулетний
biflagellate spermatozoon	ikki xivchinli spermatozoid	двухжгутиковый сперматозоид
birch	qayin	берёза
bisexual	ikki jinsli	двуполый
buckwheat	marjumak	гречиха
bud	kurtak	почка
C alyptra	qin, qalpoq	калиптра, колпачок
calyx	gulkosacha	чашечка
calyx petaloid	gultojbargsimon gulkosacha	венчиковидная чашечка
capitulum	savatcha	корзинка
capsule	ko'sak meva	коробочка
capsule (sporangium)	ko'sakcha (sporangiy)	коробочка (спорангий)
carnivorous	hasharotxo'r o'simliklar	насекомоядные растения
catkin	kuchala (to'pgul)	серёжка

cedar
 cell
 central trunk
 chlorophyllous cells
 class
 classification
 closely packed
 club moss
 compound inflorescence
 compound leaf
 compound spike
 conduction (root function)
 cone, strobilus
 conifers
 corolla
 cortex
 cottonwood
 cotyledon
 cover slip
 cypress

Dead cells
 deciduous
 dehiscent
 dense racemes

kedr
 hujayra
 markaziy tana
 xlorofilli hujayralar
 ajdod
 klassifikatsiya, tasnif
 zich joylashgan
 plaun
 murakkab to'pgul
 murakkab barg
 murakkab boshqoq
 o'tkazuvchi
 g'udda
 ignabargli (daraxtlar)
 gultoj
 po'st
 terak
 urug'palla
 qoplag'ich oyna
 kiparis

o'lik hujayralar
 bargini to'kuvchi
 chatnaydigan
 tig'iz (zich) shingil

кедр
 клетка
 центральный ствол
 хлорофиллоносные клетки
 класс
 классификация
 плотно расположенные
 плаун
 сложное соцветие
 лист сложный
 сложный колос
 проводящий
 шишка
 хвоя (хвойные деревья)
 венчик
 корка
 тополь
 семядоля
 покровное стекло
 кипарис

мертвые клетки
 листопадные растения
 нераскрывающиеся плоды
 плотный кисть

dentate	tishchali	зубчатый
dichotomously branched veins	dixotomik shoxlangan tomirlar	дихотомически разветвленные жилки
dichotomously veined blade	dixotomik tomirlangan barg yaprog‘i	дихотомически жилкованная пластинка
dicots	ikki urug‘pallali o‘simliklar	двусемядольные растения
difference, dissimilarity	farq	различие
dioecious	ikki uyli o‘simlik	двудомный
diploid	diploid	диплоид
dissolved minerals or salts	erigan minerallar yoki tuzlar	растворенные минералы или соли
distinctive feature	o‘ziga xos xususiyat	своеобразная особенность
distribution	tarqalish	распространение
division	bo‘lim	отдел
Douglas-fir	soxta tsuga, Duglas pixtasi	псевдотсуга, дугласова пихта
drupe	danakli meva	костянка
dry fruit	quruq meva	сухие плоды
E laters (spring-shaped)	elatera iplari (prujinasimon)	элатеры (пружинистые)
embryo	murtak	зародыш
embryo coiled	o‘rama murtak yoki murtagi o‘ralgan	намотанный зародыш
enclosed seed plants	yopiq urug‘li o‘simliklar	покрытосеменные растения
eternal young	doimiy yosh	вечно молодой
evergreen	doimiy yashil	вечнозеленый
evolution	evolyutsiya	эволюция
exchange of moisture	nam almashinuvi	влагообмен
exposed seed plants	ochiq urug‘li o‘simliklar	голосеменные растения

extinct	yo‘qolgan	исчезнувший
extinct species	yo‘qolgan turlar	исчезнувшие виды
F amily	oila	семейство
fan-shaped	elpig‘ichsimon	вееровидный
fern	qirqquloq	папоротник
fertilization	urug‘lanish	оплодотворение
fibrous root	popuk ildiz	мочковатый корень
filament	changchi ipi	тычиночная нить
fir	oqqarag‘ay, pixta	пихта
fire preventing source	yong‘inga qarshi vosita	противопожарное средство
fire-adapted	yong‘inga moslashgan	адаптированность к пожару
flagellum (<i>pl. flagella</i>)	xivchin	жгутик
flatining, softening, polishing	silliqlash	полирование
fleshy fruits	ho‘l meva	сочные плоды
floral axis, receptacle	gul o‘rni	цветоложе
flower	gul	цветок
flowering	gullash	цветение
follicle	bargak	листовка
forest	o‘rmon	лес
fossil species	qazilma turlar	ископаемые виды
fruit	meva	плод
fruitining	meva tugish	плодоношение
fungal diseases	zamburug‘ kasalliklari	грибковое заболевание
fungal hyphae	zamburug‘ iplari	грибковые гифы

fused (petals)

Gametophyte

gemma cups

generative organs

genus

germination

gialin cells

glabrous leaf

glaucous wax bloom

grain or caryopsis

growing point

growth rings

guard cells

gynoecium

Habitat

half-inferior ovary

halophytic

haploid

hard wood

head

heavy metals

hemlock

herb

qo‘shilgan (gultojbarg)

gametofit

chiqaruvchi (ajratuvchi) kurtaklar

generativ organlar

turkum

urug‘ning unishi

gialin hujayralar

silliq barg

ko‘kintir mum qavati

don meva

o‘shish nuqtasi

yillik halqa

qamrovchi, lab hujayralari

ginetsey

yashash muhiti

yarim ostki tuguncha

galofit

gaploid

qattiq yog‘och

boshcha

og‘ir metallar

tsuga

o‘t

сращённый (лепесток)

гаметофит

выводящие почки

генеративные органы

род

прораствание семян

гиалиновые клетки

лист гладкий

сизо восковый налёт

зерновка

точка роста

годовые кольца

закрывающие клетки

гинецей

местообитание

полунижняя завязь

галофитные

гаплоид

прочная древесина

головка

тяжелые металлы

тсуга

травы

hesperidium	gesperidiy	гесперидий
hibiscus	gibiskus	гибискус
high latitude	yuqori kenglik	высокая широта
higher plants	yuksak o'simliklar	высшие растения
hollow stem	somon poya	соломина
horsetail	qirqbo'g'im	хвощ
Inconspicuous	ko'zga ko'rinmas, juda kichkina	невидимый
indehiscent	chatnamaydigan	раскрывающиеся плоды
indusium	induzium	индузиум
inferior ovary	ostki tuguncha	нижний завязь
integument	integument	интегумент
Juniper	archa	можжевильник
jurassic Period	yura davri	юрский период
Lanceolate	nashtarsimon	ланцетный
land	quruqlik	суша
lateral root	yon ildiz	боковой корень
lattitudinal (cross) section	ko'ndalang kesim	поперечный срез
leaf (<i>pl.</i> : leaves)	barg	лист
leaf arrangement	bargning joylashuvi	листорасположение
leaf collar	barg bo'g'imi	листовой узел
leaf margins, leaf edge	barg qirrasi, barg cheti	край листовой пластинки

leaf vein	barg tomiri	жилка листа
leaf venation	bargning tomirlanishi	жилкование листа
leafless	bargsiz	безлиственный
leaflet, blade	barg plastinkasi, yaproq	листовая пластинка
legume	dukkak meva	боб
life cycle	taraqqiyot sikli	цикл развитие
lignification	lignifikatsiya, yog‘ochlashish	лигнификация, одревеснение
longitudinal section	uzunasiga kesim	продольный срез
loosely packed	tarqoq joylashgan	рыхло расположенные
M ajority	ko‘pchilik	большинство
mallow	gulxayri	мальва
maturity	etilganlik	зрелость
mechanical tissue	mexanik to‘qima	механическая ткань
meiosis	reduksion bo‘linish	редукционное деление
microphyllous	mikrofiliya	микрофилия
micropyle	mikropile	микропиле
microscope objective	mikroskop ob‘ektivi	объектив микроскопа
microscope slide	buyum oynasi	предметное стекло
midrib (leaf)	bargning markaziy tomiri	центральная жилка листа
moist	nam	увлажненный
monocots	bir urug‘pallali o‘simliklar	односемядольные растения
monoecious	bir uyli o‘simlik	однодомный
monoflagellate spermatozoon	bir xivchinli spermatozoid	одножгутиковый сперматозоид
mosses	yo‘sin, mox	мхи

multiflagellate spermatozoon	ko‘p xivchinli spermatozoid	многожгутиковый сперматозоид
multinucula	to‘pyong‘oq	многоорешок
multiple	to‘p meva	соплодие
mustard	xantal	горчица
mycorrhiza	mikoriza	микориза
N etlike veins	panjasimon (to‘rsimon)	сетчатое
nitrogen fixing	azot to‘plovchi	азотофиксирующий
node	novda bo‘g‘ini	узел побега
non-vascular, lower plants	tuban o‘simliklar	низшие растения
northern hemisphere	shimoliy yarim shar	северное полушарие
northern pole	shimoliy qutb	северный полюс
nucellus	nutsellus	нуцеллус
nut	yong‘oq	орех
nutlet	yong‘oqcha	орешек
O ak	eman	дуб
oblique (leaf)	assimetrik (barg)	асимметрический (лист)
obsolete	eskirgan	устаревший
operculum	qopqoqcha	крышечка
opposite	qarama-qarshi	супротивное
order	qabila	порядок
organic substances	organik moddalar	органические вещества
ovary	tuguncha	завязь
ovule	urug‘kurtak	семяпочка

ovum, egg cells
oxygen

Palisade mesophyll
panicle
panicle-type inflorescence
parallel veins
paraphyses (in Funaria)
peat
peat pots
pedicel, flower stalk
peltate
pepo
perennial
perianth
persitome teeth
petal
petiolate leaf
petiole
phloem
pine
pinnate
pinnatifid
pipette
pistil

tuxumhujayra
kislorod

ustunsimon (palisad) parenxima
ro‘vak
ro‘vaksimon to‘pgul
parallel
parafiza iplari
torf
torf tuvaklari
gul bandi
qalqonsimon
qovoq meva
ko‘p yillik
gulqo‘rg‘on
peristom tishlari
gultojbarg
bandli barg
barg bandi
floema
qarag‘ay
patsimon
patsimon qirqilgan
suv tomizgich
urug‘chi

яйцеклетка
кислород

столбчатая паренхима
метёлка
метёлковидное соцветие
параллельное
парафиза
торф
торфяной горшок
цветоножка
щитовидный
тыква
многолетний
околоцветник
зубцевидные перистомы
лепестки
лист черешковый
черешок листа
флоэма
сосна
перистый
перисторассеченный
пипетка
пестик

pistillate flower	urug‘chili gul	пестичный цветок
pith	o‘zak	сердцевина
placenta	platsenta	плацента
plant body	o‘simlik tanasi	ствол растения
plant kingdom	o‘simliklar dunyosi, olami	царство растений
plant origin	o‘simlikning kelib chiqishi	происхождение растений
pollen	chang	пыльца
pollen-producing structure	chang hosil qiluvchi struktura	пыльцеформирующая структура
pollination	changlanish	опыление
pollution	ifloslanish	загрязнение
polypetalous	ko‘p gultojbargli	многолепестный
pome	olma	яблоко
prepared microscope slides	tayyor mikroskopik preparatlar	готовые микроскопические препараты
primary vascular cells	birlamchi o‘tkazuvchi (naychali) hujayralar	первичные сосудистые клетки
primary xylem/phloem	birlamchi ksilema/floyema	первичная ксилема/флоэма
prothallus	o‘simta	заросток
pubescent	tukli	опушённый
Raceme	shingil	кисть
rachis	boshqoq o‘qi	стержень колоса
rainfall	yog‘ingarchilik	осадки
remote portion	eng chekka qismi	самая отдаленная часть
representative	vakil	представитель

reproduction	ko‘payish	размножение
rhizome	ildizpoya	корневище, подземный стебель
root	ildiz	корень
root cap	ildiz qini	корневой чехлик
root hairs	ildiz tuklari	корневые волоски
root neck (...crown, ... collar)	ildiz bo‘yni	корневая шейка
root system	ildiz tizimi	корневая система
rootstock	ildizpoya	корневище
Samara	qanotchali	крылатка
scale-like leaves, squamata leaves	qipiqchali, tangachali barg	чушуйчатые листья
secondary growth	ikkilamchi o‘sish	вторичный рост
seed	urug‘	семена
seedling	o‘simta	проросток
sepal	gulkosachabarg	чашелистник
serrate	arrasimon	пильчатый
sessile leaf	bandsiz barg, o‘troq barg	лист сидячий
seta (of capsule)	ko‘sakcha bandi	нитка коробочки
sexual phase (<i>gametophyte</i>)	jinsiy bo‘g‘in (<i>gametofit</i>)	половая фаза (<i>гаметофит</i>)
shady	soya	тень
shell-like shape	chig‘anoqsimon ko‘rinish	раковинообразный вид
shoot system	er ustki qismi	надземная часть
shrub	buta	кустарник
sieve tubes	elaksimon (to‘rsimon) naycha	ситовидные трубки
silicon	kremniy	кремний

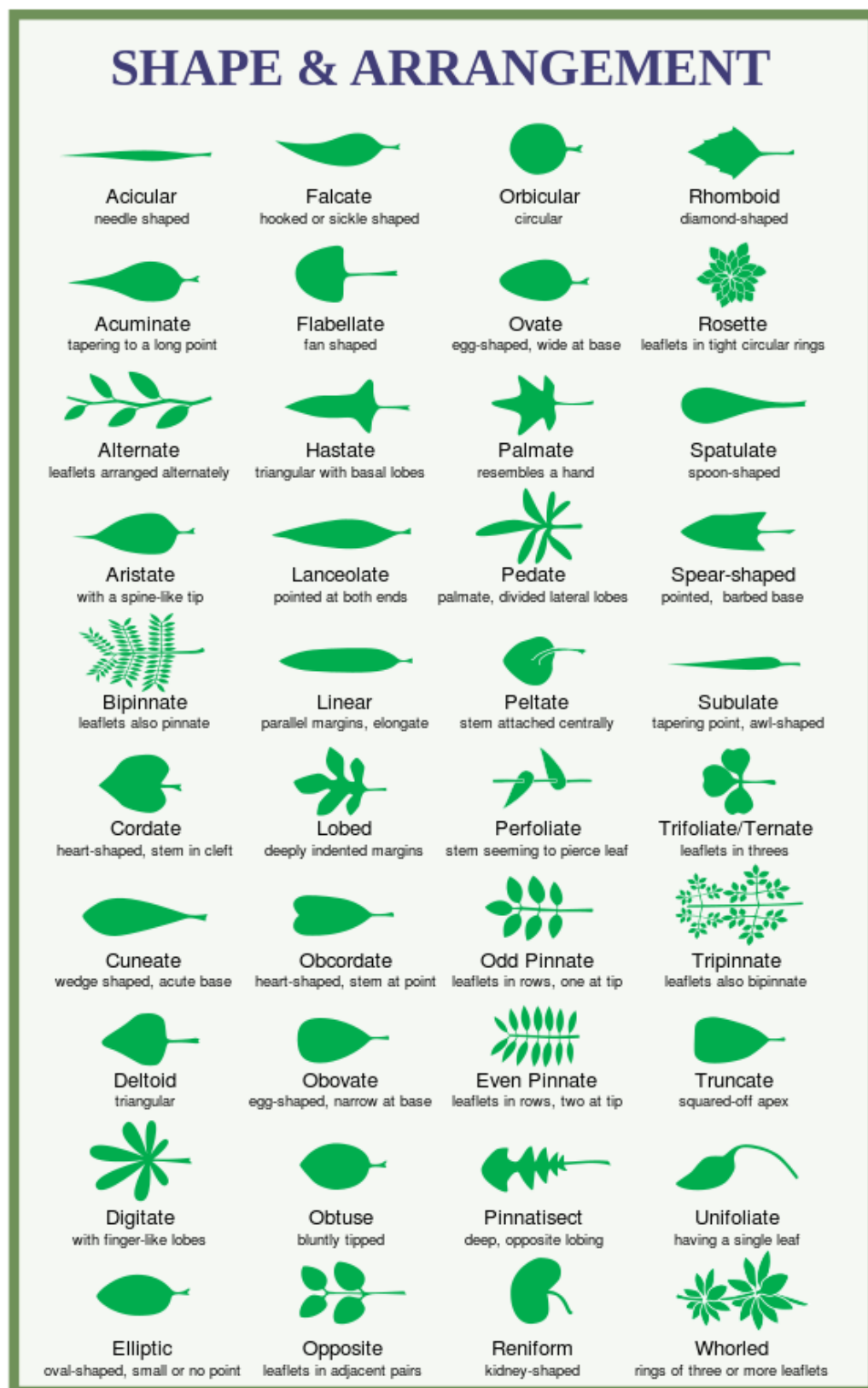
silique	qo‘zoq meva	стручок
silmlarity	o‘xshashlik	сходство
simple inflorescence	oddiy to‘pgul	простое соцветие
simple leaf	oddiy barg	лист простой
smartweed	suv qalampiri	горец
solitary	yakka gul	одиночные цветы
sorus (<i>pl. sori</i>)	sorus	сорус
speciemen	namuna	образец
species	tur	вид
spermatozoon, spermatozoid	spermatozoid	сперматозоид
spike	boshog	колос
spikelet	boshogcha	колосок
spinach	ismaloq	шпинат
spongy mesophyll	bulutsimon (g‘ovak) parenxima	губчатая паренхима
sporangium	sporangiy	спорангий
spore	spora	спора
sporogonium	sporogon	спорогон
sporoous capsule	spora boshog‘i	споровый колосок
sporophyll	sporofill	спорофилл
sporophyte	sporofit	спорофит
spruce	qoraqarag‘ay	ель
stalk (on thallus)	tirgovuch (tallomda)	подставка
stamen	changchi	тычинка
staminate flower	changchili gul	тычиночный цветок
starch	kraxmal	крахмал

stellate hair, trichome	trixoma, tuk	трихома
stem	poya	стебель
stigma	tumshuqcha	рыльце
stipule	yonbargcha	прилистник
stomata	og‘izcha	устьица
style	ustuncha	столбик
subclass	kichik ajdod	подкласс
sunlight	quyosh nuri	солнечные лучи
superior ovary	ustki tuguncha	верхний завязь
supportive (stem function)	tayanch (poyaning vazifasi)	опорный
swamp, peatland, wetland	botqoqlik	болото
swollen nodes	bo‘rtgan bo‘g‘in	раздутая пазуха
sympetalous	qo‘shilgan gultojbarg	спайнолепестный
synangium	sinangiy	синангий
syncarpous	sinkarp, ko‘p chanoqli	синкарпный
systematics	sistematika	систематика
Tap root	o‘q ildiz	корень стержневой
tetraquetrous, tetrahedral	to‘rt qirrali	четырёхгранный
thallus (<i>pl. thalli</i>)	tallom	таллом
tillering	shonalash	кущение
to disperse	tarqalmoq	распространяться
tracheids	traxeid	трахеид
trifoliate	uchtalik (barg)	трехлистный
triquetrous, trihedral	uch qirrali	трехгранный

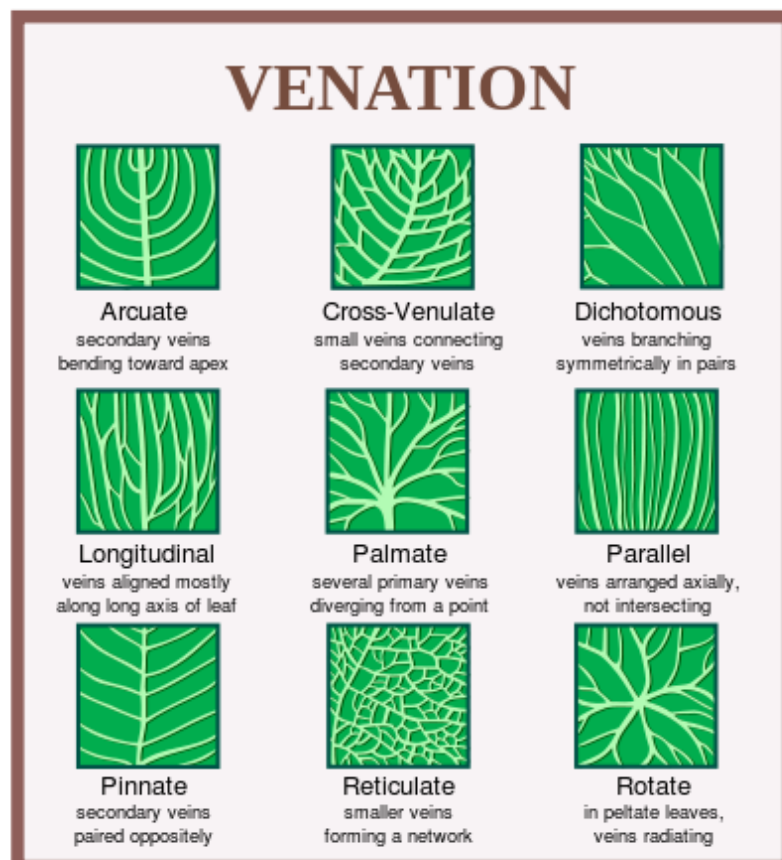
true fruit	chin meva	плод настоящий
true mosses	poyabargli yo'sinlar	листочекбельные мхи
tufts of hairs	tuklar (trixoma) tutami yoki bog'lami	пучок трихомы
tweezer	pinset	пинцет
U ltraviolet light	ultrabinafsha nurlar	ультрафиолетовые лучи
umbel	soyabon	зонтик
unisexual	bir jinsli	однополый
upper epidermis	yuqori epiderma	верхняя эпидерма
upright (stem)	tik (poya)	стоячий (побег)
V alve	klapan	клапан
vascular bundles	o'tkazuvchi boylamlar	проводящие пучки
vascular plants	tomirli (naychali) o'simliklar	сосудистые растения
vascular system	o'tkazuvchi nay-tolali bog'lamlar	сосудисто-волокнистые пучки
vascular tissues	o'tkazuvchi to'qimalar	проводящие ткани
vegetative organs	vegetativ organlar	вегетативные органы
vessel, trachea	traxeya (nay)	трахея
W ater holding capacity	suv saqlash xususiyati	водоудерживающая способность
waxy cuticle	mum pardali kutikula	кутикула с восковым налетом
weed	begona o't	сорное растение, сорняк
well-defined	yaxshi farqlanadigan	хорошо различаемый
well-developed embryos	yaxshi shakllangan murtak	хорошо развитый зародыш

whorled	halqasimon	мутовчатое
willow	tol	ива
wind-pollinated	shamol yordamida changlanuvchi	ветроопыляемые
woody plant	daraxtchil o'simliklar	древесные растения
xeric environments	qurg'oqchil muhit	засушливая среда
xylem	ksilema	ксилема
Y ew	tis	тис
yield	hosil	урожай
Z ygomorphous, zygomorphic	zigomorf	зигоморф
zygote	zigota	зигота

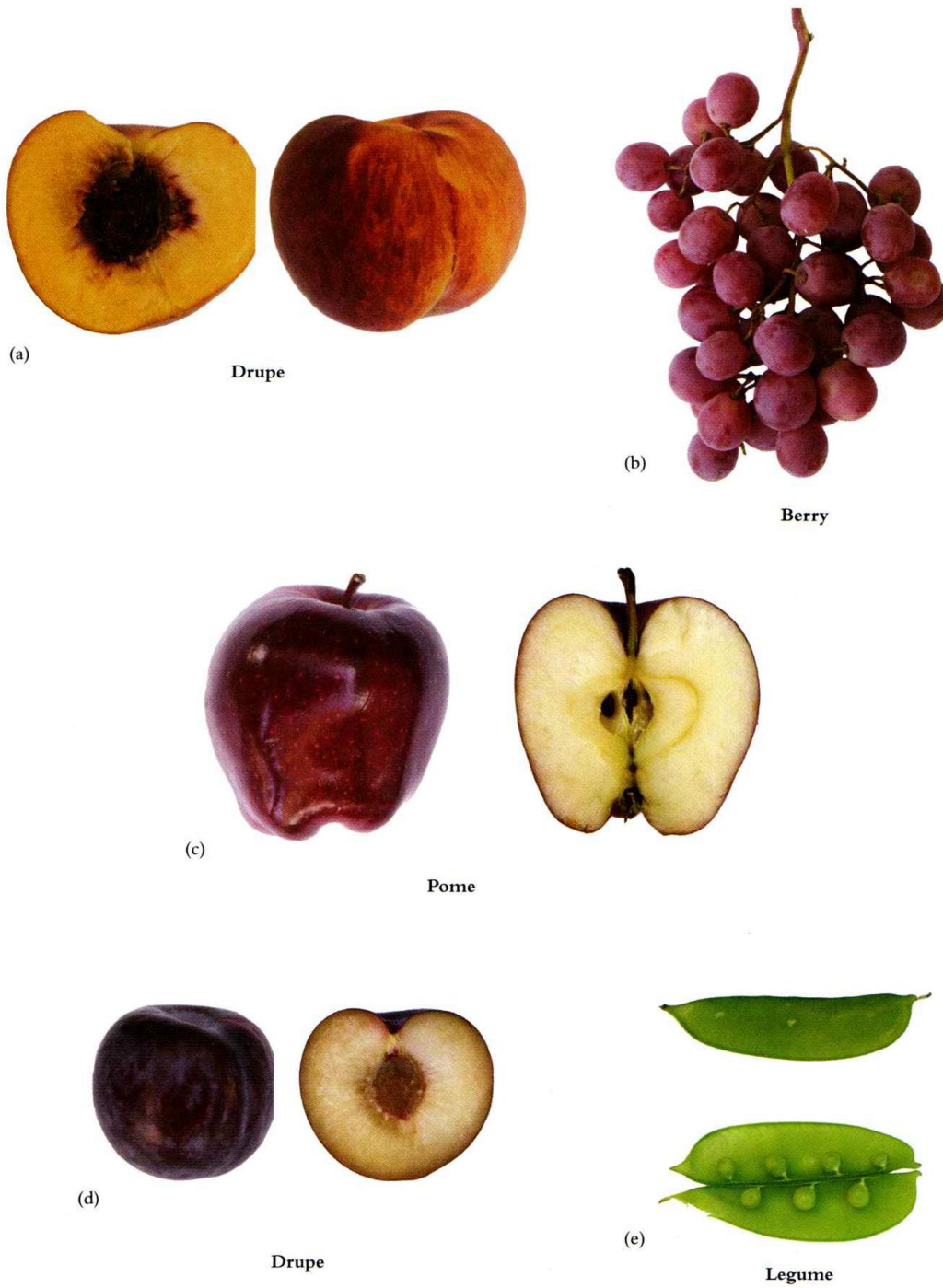
1. Leaf shapes and arrangements



2. Leaf margins and venation



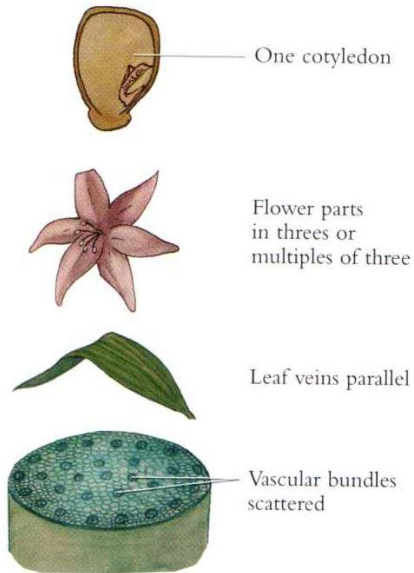
3. Fruit types



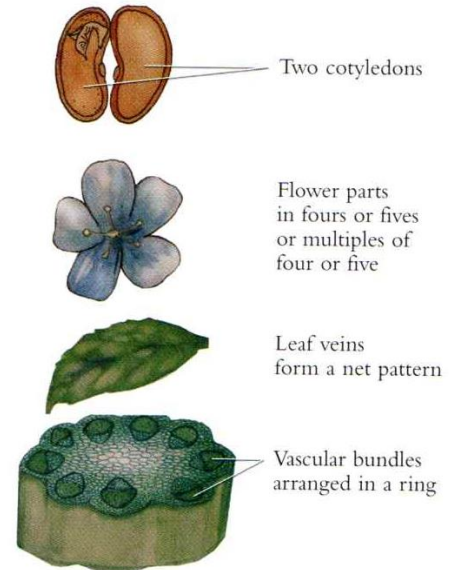
Source: Van De Graaff et al., 2004

4. Some characteristics of monocots and dicots

Class Monocotyledonae (monocots)



Class Dicotyledonae (dicots)



Some examples of monocots



Wheat



Corn



Cattail



Iris

Some examples of dicots



Water lily



Columbine



Rose



Sunflower

Source: Van De Graaff et al., 2004

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**T.F.Rajabov, M.G.Nosirov,
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